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[314] Udry, J. R., Bearman, P. S., and Harris, K. M., National Longitudinal Study of Adolescent Health. This work uses data from Add Health, a program project designed by J. Richard Udry, Peter S. Bearman, and Kathleen Mullan Harris, and funded by a grant P01-HD31921 from the Eunice Kennedy Shriver National Institute of Child Health and Human Development, with cooperative funding from 23 other federal agencies and foundations. Special acknowledgment is due Ronald R. Rindfuss and Barbara Entwisle for assistance in the original design. Persons interested in obtaining data files from Add Health should contact Add Health, Carolina Population Center, 123 W. Franklin Street, Chapel Hill, NC 27516-2524 (addhealth@unc.edu). No direct support was received from grant P01- HD31921 for this analysis.

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SMTP (Simple Mail Transfer Protocol)

snowball sampling

and eigenvector centrality

biases

network reconstruction from

social network

actors

AddHealth study

animals

animosity

archival data

assortative mixing

average degree

biases in

bipartite

blogs

brokers

clustering coefficient

community structure

degree

degree distribution

directed networks

direct observation

disassortative mixing

disease spread

edges

email network

empirical measurements of

Facebook

film actor network

friendship network

geodesic distances

groups in

in-degree

instant messaging

intermarriage network

karate club network

LinkedIn

LiveJournal

mean degree

medical doctors

MySpace

online

origin of the term

out-degree

percolation on

personal networks

questionnaires

respondent-driven sampling

rumor spreading

schoolchildren

shortest distances

Southern Women Study

sparse networks

third-party records

time-resolved

triads in

Usenet

weblogs

social network analysis

Social Science Citation Index

sociogram

sociometric network studies

sociometric superstars

sociometry

software call graph

software class network

software packages

Graphviz

GTL

igraph

InFlow

JUNG

LEDA/AGD

Mathematica

Matlab

Netminer

network of

Network Workbench

NetworkX

Pajek

UCINET

Visone

web crawlers

yEd

source food web

Southern Women Study

affiliation network

bipartite network

picture of

spanning percolation cluster

spanning tree

sparse network

adjacency matrix of

algorithms on

breadth-first search on

eigenvectors of

exponential random graphs

friendship networks

Internet

running time of algorithms

social networks

World Wide Web

spectral gap

spectral partitioning

algebraic connectivity

community detection

examples

Laplacian

modularity matrix

running time

spectrum

adjacency matrix

Laplacian

spontaneous symmetry breaking

stability analysis

stability condition

stability function

star graph

betweenness centrality

transportation network

static web

Stirling's approximation

stratified network

Strauss model

stretched exponential

strongly connected component

acyclic networks

citation networks

cycles in

directed random graphs

giant

largest

small

World Wide Web

structural balance

structural equivalence

structural holes

subnet

class A and B

class C

Internet representation

superhub

supernode

superpeer

surveys

design of

fixed choice

free choice

General Social Survey

name generators

respondent-driven sampling

telephone surveys

susceptible-infected model, *see* SI model

susceptible-infected-recovered model, *see* SIR model

susceptible-infected-susceptible model, *see* SIS model

susceptible state

symmetric fixed point

symmetry breaking

synchronization

tandem affinity purification

TCP (Transmission Control Protocol)

technological networks

airline network

assortative mixing

disassortative mixing

distribution networks

empirical measurements of

gas pipelines

Internet

oil pipelines

power grid

rail network

road network

statistics

telephone network

transportation networks

telephone call graph

telephone company

telephone exchange

local

long distance

telephone network

circuit switched

geography

long-distance offices

packet switched

schematic representation

toll-switching offices

telephone surveys

Teoma

thesaurus network

third-order phase transition

threshold

appearance of giant component

epidemic

percolation

tie

time complexity

time-resolved social network

toll-switching office

traceroute

train network, *see* rail network

transcription factor

transfer RNA

transitivity

cliques

directed networks

exponential random graphs

food webs

Internet

partial

perfect

Poisson random graph

random graphs

small-world model

social networks

Strauss model

World Wide Web

transmission probability

transmission rate

transportation network

airline network

maintenance costs

optimization

rail network

road network

star graph

tradeoffs

trapezium rule, *see* trapezoidal rule

trapezoidal rule

tree

absence of loops in

adjacency tree

and configuration model

and random graph

AVL tree

Bethe lattice

betweenness centrality

Cayley tree

clustering coefficient

data structure

dendrogram

hierarchical network model

in-components

leaf nodes

minimum spanning tree

number of edges

percolation on

planarity

randomized

river networks

rooted

root node

small components

tree data structure

adding an element to

AVL tree

balanced

binary heap

child node

deleting an element in

depth

enumerating elements of

Euler tour

finding an element in

leaf node

parent node

pivot

rebalancing

root node

triad

closed

signed networks

social networks

stable

unstable

triadic closure

triangles

and clustering coefficient

random graphs

small-world model

Strauss model

triangular lattice

triangular lattice

triangular matrix

tricomponent

non-contiguous

tripartite network

trophic level

trophic species

trust network

two-hybrid screen

two-mode network

two-star

two-star model

coexistence region

continuous phase transition

edge probability

Hamiltonian

mean-field theory

phase diagram

phase transition

problems with

spontaneous symmetry breaking

UCINET (software package)

UDP (User Datagram Protocol)

undirected network

adjacency list

adjacency matrix

average degree

betweenness centrality

clustering coefficient

components

connectivity

degree

degree distribution

degree sequence

eigenvector centrality

excess degree distribution

exponential random graph

friendship networks

from directed network

Katz centrality

***k*-components**

loops in

metabolic networks

paths on

self-edges

sexual contact networks

social networks

transitivity

URL

Usenet

vaccination

acquaintance immunization

non-uniform

percolation theory

scale-free network

targeted

valued network, *see* weighted network

vertex connectivity

vertex copying

biological networks

citation networks

gene duplication

metabolic networks

model

protein-protein interaction networks

vertex copying model

citation networks

connection to Price model

exponent

master equation

power-law degree distribution

vertex cut set

vertex-disjoint paths, *see* vertex-independent paths

vertex-independent paths

algorithm for

vertex percolation, *see* site percolation

vertices

age of

centrality

copying

cut set

examples of

food webs

groups of

high degree

highest degree

importance of

Internet

metabolic networks

percolation on

removal

similarity

social networks

values on

World Wide Web

virus, computer

Visone (software package)

visualization

acyclic networks

software

water supply network

Watts-Strogatz model

weakly connected component

directed random graph

giant

largest

web crawler

biases

breadth-first search

for citations

operation of

robots.txt

software

web search

web link, *see* hyperlink

weblog network

web

dynamic

indexes of

ranking of

static

web search

anchor text

Ask.com

breadth-first search

Google

PageRank

Teoma

web crawlers

website

weighted edge

negative weights

weighted network

adjacency matrix

betweenness centrality

bibliographic coupling networks

bipartite

cocitation networks

connection to multigraphs

food webs

geodesic paths

max-flow/min-cut theorem

maximum flow

minimum cut

one-mode projections

recommender networks

shortest paths on

shortest path trees

Westlaw

W-function

wolves, social network

word network

World Wide Web

as distributed database

assortative mixing

Barabási-Albert model

blogs

bow tie diagram

clustering coefficient

co-links

community structure

components

cumulative degree distribution

degree distribution

directed network

disappearance of edges

disappearance of vertices

edges

exponent

giant in-component

giant out-component

hyperlinks

in-components

in-degree

in-degree distribution

large components

largest strongly connected component

models of

nodes

number of

out-components

out-degree

out-degree distribution

power-law degree distribution

Price model

reciprocity

scale-free network

search engine

sparse network

statistics

strongly connected components

transitivity

vertices

weblogs

web search

yeast protein-protein interaction network

yeast two-hybrid screen

yEd (software package)

Yule process

Yule, Udney

Zachary, Wayne

zeta function

generalized

incomplete

z-transform

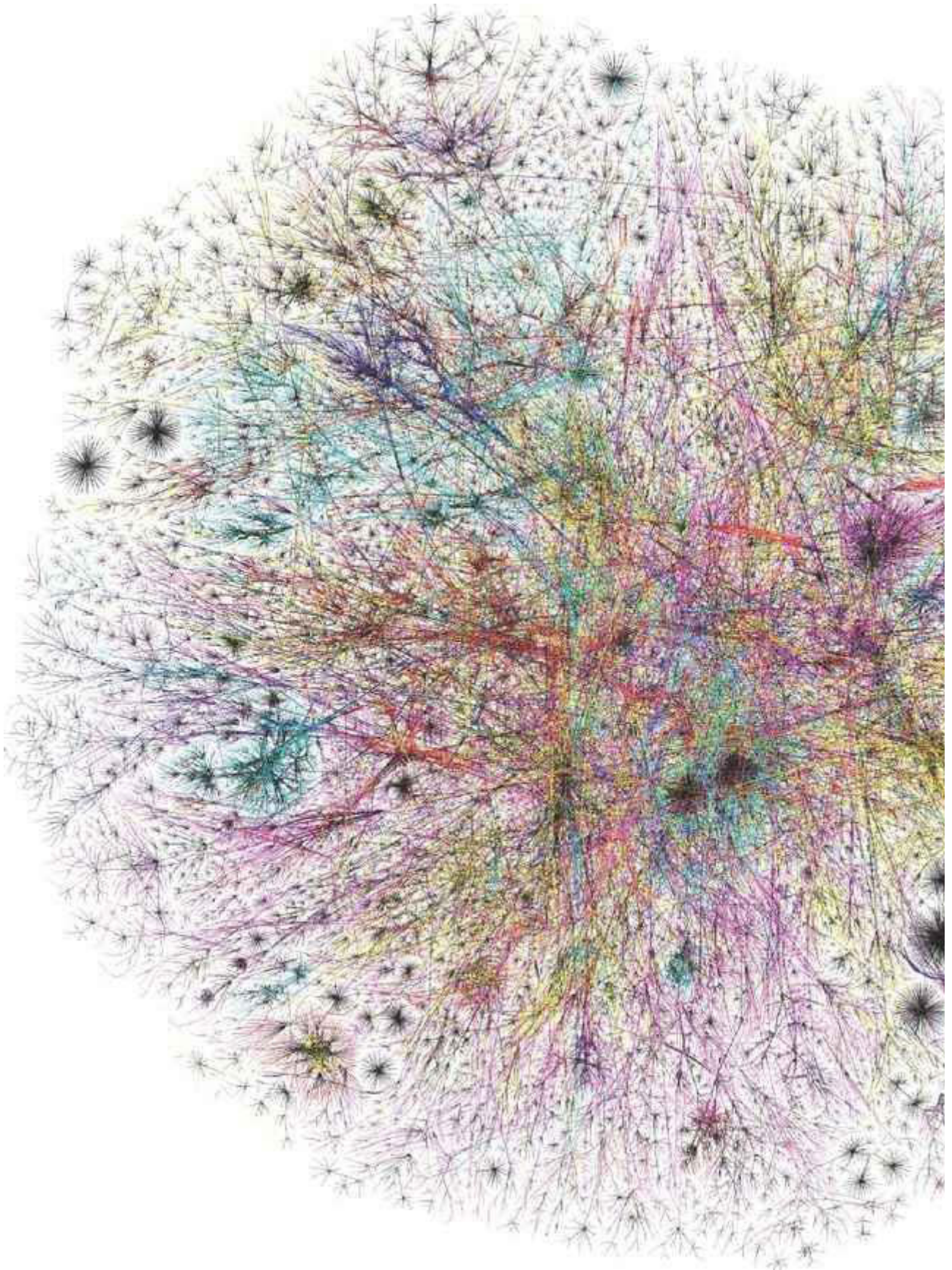


Plate I: The network structure of the Internet. The vertices in this representation of the Internet

are “class C subnets”—groups of computers with similar Internet addresses that are usually under the management of a single organization—and the connections between them represent the routes taken by Internet data packets as they hop between subnets. The geometric positions of the vertices in the picture have no special meaning they are chosen simply to give a pleasing layout and are not related, for instance, to geographic position of the vertices. The structure of the Internet is discussed in detail in Section 2.1. Figure created by the Opte Project www.opte.org. Reproduced with permission.

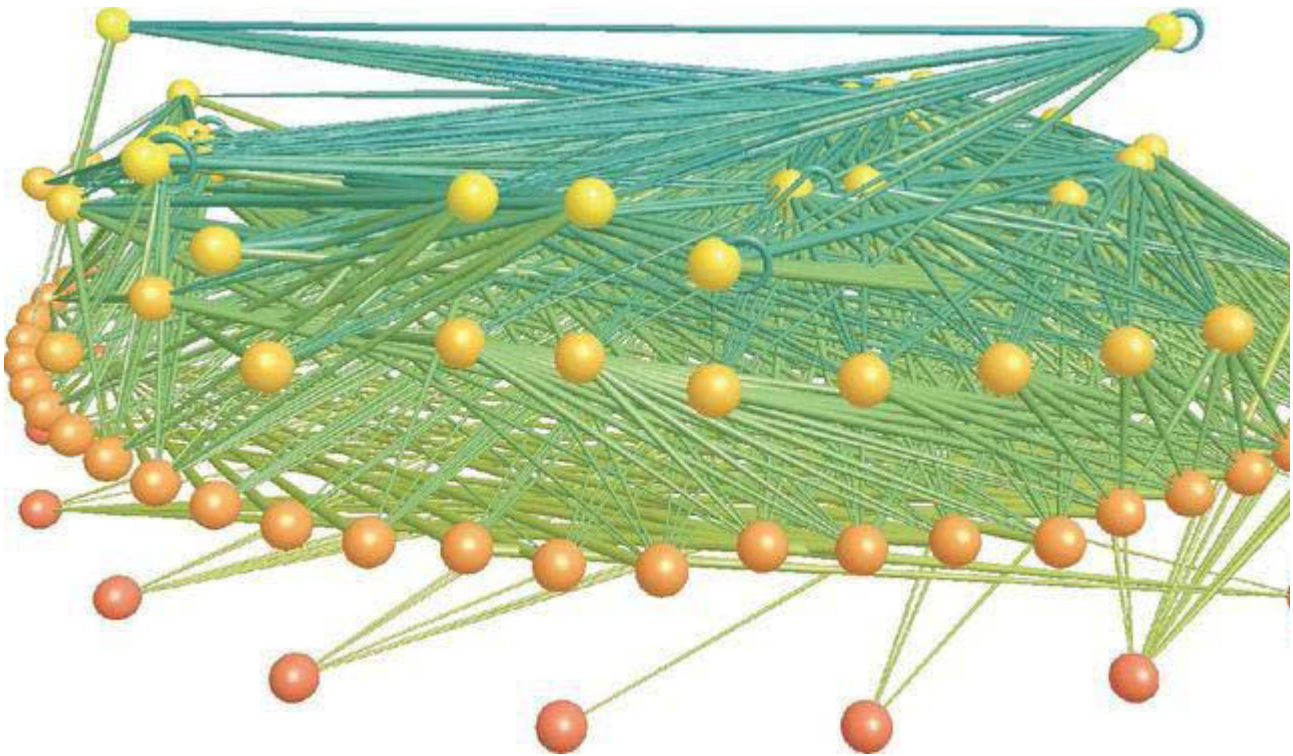


Plate II The food web of ittle ock ake Wisconsin. This elegant picture summarizes the known predatory interactions between species in a freshwater lake in the northern United States. The vertices represent the species and the edges run between predator-prey species pairs. The vertical position of the vertices represents, roughly speaking, the trophic level of the corresponding species. The figure was created by Richard Williams and Neo Martinez [210].

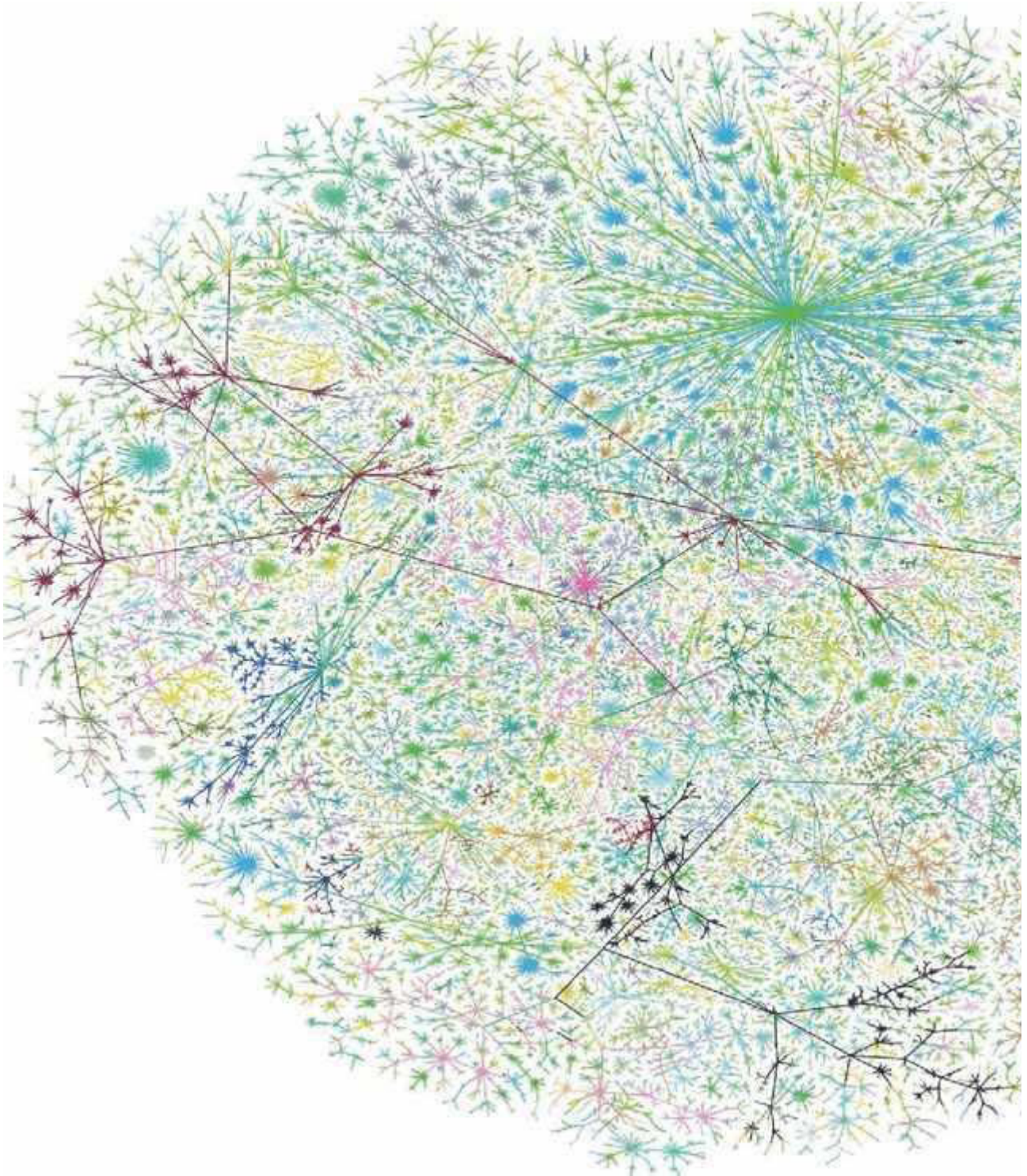


Plate III The structure of the Internet at the level of autonomous systems. The vertices in this network representation of the Internet are autonomous systems and the edges show the routes taken by data traveling between them. This figure is different from Plate I, which shows the network at the level of class C subnets. The picture was created by al Burch and Bill Cheswick. Patent(s) pending and Copyright Lumeta Corporation 2009. Reproduced with permission.

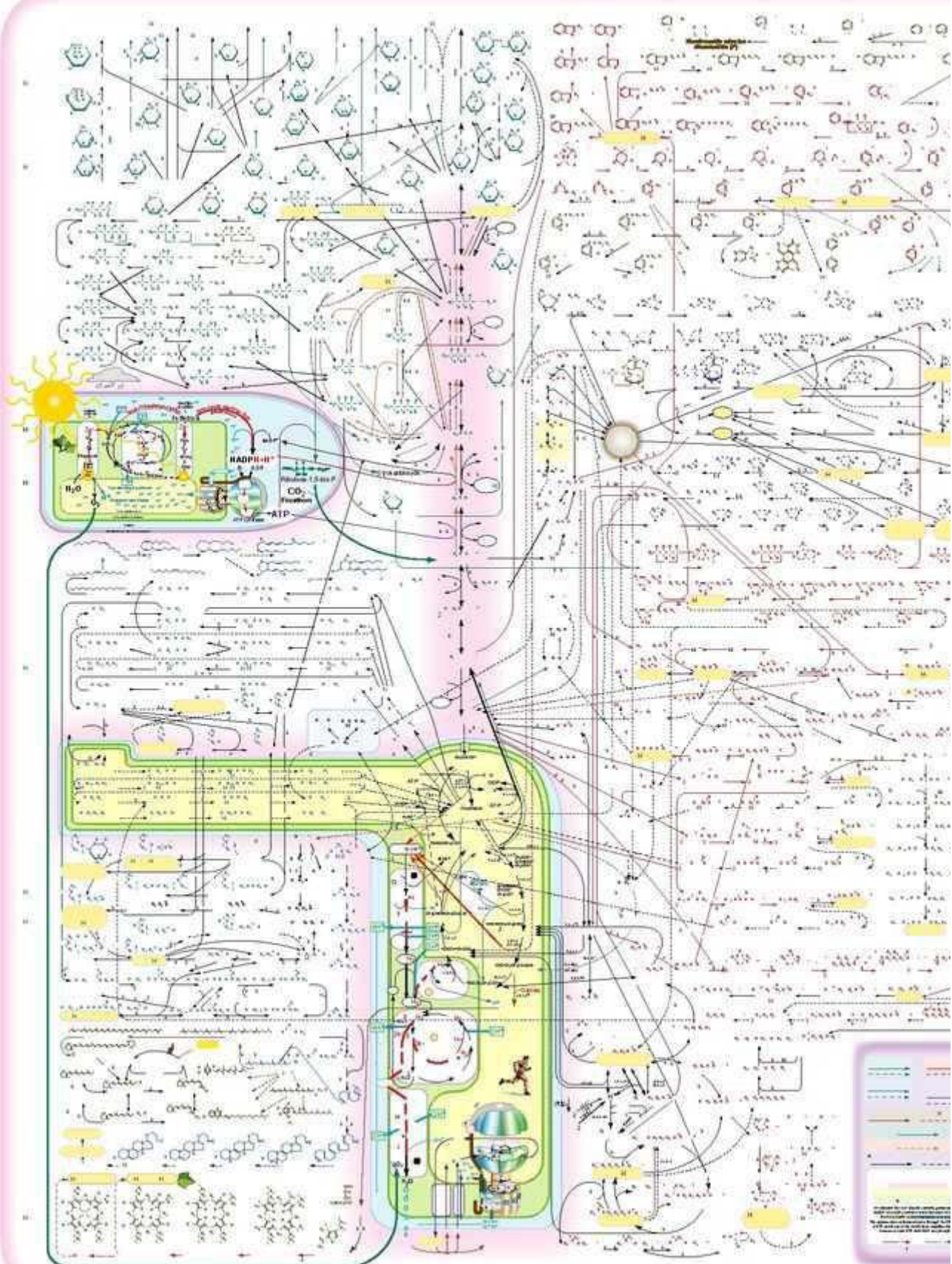


Plate I metabolic network. A wallchart showing the network formed by the major metabolic pathways. Created by Donald Nicholson. Copyright of the International Union of Biochemistry and Molecular Biology. Reproduced with permission.

1

Singular: vertex.

2

An example appears as Fig. 5.2 on page 83.

3

One must be careful when there are vertex pairs in the network that are connected by no path at all. Such issues are dealt with in Section 8.2.

4

Most of the well-known communications protocols of the Internet are themselves built on top of TCP, including HTTP (the World Wide Web), SMTP (email), and FTP (file transfer). Thus communication is a three-layer process with a user-level protocol running on top of TCP, which in turn runs on top of IP, and the user protocols automatically benefit from the error-checking features and guaranteed transmission offered by TCP. (There are lower-level transport protocols as well, such as Ethernet, PPP, and ATM, but these will not concern us.) There are however also some applications of Internet technology that do not require guaranteed transmission. Most of the common examples are streaming media, such as audio and video transmissions, voice and teleconferencing, and online games. An alternative protocol to TCP called *User Datagram Protocol* (UDP), which provides no transmission guarantees, is used in such cases.

5

We are assuming that each packet takes the same route to the destination. It is possible, but relatively rare, for different packets to take different routes, in which case the set of IP addresses returned by the traceroute procedure will not give a correct path through the network. This can happen, for instance, if congestion patterns along the route vary significantly while the procedure is being performed, causing the network to reroute packets along less congested connections. Serious Internet mapping experiments perform repeated traceroute measurements to minimize the errors introduced by effects such as these.

6

See www.netdimes.org.

7

If there were a unique best path to every vertex, then the set of paths would be precisely a tree, i.e., it would contain no loops. Because of the way routing algorithms work, however, this is not in practice always the case—two routes that originate at the same point and pass through the same vertex on the way to their final destination can still take different routes to get to that vertex, so that the set of paths can contain loops.

8

See www.routeviews.org.

9

Social networks are perhaps the main exception—in many cases people or groups of people can be considered to have reasonably well-defined geographic locations. Relatively little work has been done however on the effects of geographic distribution, perhaps because most social network studies have concentrated on populations in local neighborhoods, rather than ones spread out over significant geographic areas.

10

For most of its existence, the telephone network has connected together stationary telephones in fixed locations such as houses and offices using landlines. In the last twenty years or so fixed telephones have started to be replaced by wireless phones (“mobile phones” or “cell phones”), but

it is important to realize that even calls made on wireless phones are still primarily carried over the traditional landline telephone network. The signal from a wireless phone makes the first step of its journey wirelessly to a nearby transmission tower, but from there it travels over ordinary phone lines. Thus, while the advent of wireless phones has had an extraordinary impact on society, it has had rather less impact on the nature of the telephone network.

[11](#)

A junior high school in the United States is a school for children aged approximately 12 to 14 years.

[12](#)

One can, by asking both, make some estimate of the accuracy of the survey. If individuals' responses disagree too often, it is a clear sign that the reliability of the responses is poor.

[13](#)

See www.cpc.unc.edu/projects/addhealth.

[14](#)

Such networks are also called *egocentric* networks, although this term, which has its origins in social science and psychology, has taken on a different lay meaning which prompts us to avoid its use here.

[15](#)

This can be done, for example, by *random-digit dialing*, the practice of calling random telephone numbers in the target area and surveying those who answer.

[16](#)

Some care must be taken in the selection of the names, since the frequency of occurrence of names varies considerably, both from name to name, and geographically and culturally.

[17](#)

In American universities a "fraternity" is a semi-independent boarding house for male students.

[18](#)

Technically, the public key can be used to decrypt the message, but the calculation involved is extraordinarily complex and would take years or even centuries of effort on the fastest modern computers. For practical purposes, therefore, one can only decrypt the message if one has the private key.

[19](#)

In practice it is a little more complicated than this. Asymmetric cyphers are computationally demanding to implement, far more so than the traditional (but less secure) symmetric cyphers in which the same key is used by both parties. To reduce demands on computer time, therefore, one usually uses the asymmetric cypher only to transmit from one party to the other a key for use in a symmetric cypher, and then the symmetric cypher, with that key, is used for all subsequent communications. In this way one benefits from the security of public-key cryptography without the computational overhead. For our purposes in this section, however, this is just a technical detail.

[20](#)

Again, this is not completely true. One can encode a message using the public key that will decode with the same key, but again the calculations necessary to do this are extraordinarily lengthy, much lengthier than those using the private key, and hence for practical purposes only the person with the private key could have created the encrypted message.

[21](#)

Milgram was particularly influenced in his work by a mathematical paper by Pool and Kochen

[270] that dealt with the small-world phenomenon and had circulated in preprint form in the social science community for some years when Milgram started thinking about the problem, although the paper was not officially published until many years later.

[22](#)

In fact Milgram conducted several sets of small-world experiments. The one described here is the first and most famous, but there were others [186, 311].

[23](#)

The phrase “six degrees of separation” did not appear in Milgram’s writing. It is more recent and comes from the title of a popular Broadway play by John Guare [149], later made into a film, in which the lead character discusses Milgram’s work.

[24](#)

Furthermore, it appears that some of the initial recipients may have been selected not at random but by advertising for volunteers in the local newspaper [181], a procedure unlikely to produce a truly random sample of the population.

[25](#)

Also sometimes called “INDEX” experiments, which is an abbreviation for “informantdefined experiment.”

[26](#)

In snowball sampling the sample size grows exponentially with the number of sampling waves and hence one typically only performs a logarithmic number of waves, which is not enough for the sampling process to reach equilibrium. In random walk sampling the sample size grows only linearly.

[27](#)

This is only the number of reachable static pages. The number of unreachable pages is difficult to estimate, and dynamic pages (see later) are essentially infinite in number, although this may not be a very meaningful statement since these pages don’t exist until someone asks for them.

[28](#)

Which web pages a crawler finds does depend on where the crawl starts. A crawler *can* find a web page with no incoming links, for instance, if (and only if) it starts at that page. In practice, however, the starting point has remarkably little effect on what a crawler finds, since most of what is found consists of the giant out-component mentioned above, whose content does not depend on the starting point.

[29](#)

Indeed, academic studies of the Web within the information sciences sometimes refer to hyperlinks as “citations,” a nomenclature that emphasizes the close similarities.

[30](#)

On rare occasions it occurs that an author or authors will publish two papers simultaneously in the same volume of a journal and, with the help of the printers, arrange for each paper to cite the other, creating a cycle of length two in the network. Thus, the citation network is not strictly acyclic, having a small number of short cycles scattered about it.

[31](#)

And it’s been cited one more time now.

[32](#)

Westlaw is owned and operated by Thomson Reuters, the same company that owns the Science

Citation Index.

[33](#)

The Napster name was later bought up by the recording industry and is now the name of a legitimate online music service, although one that does not make use of peer-to-peer technology.

[34](#)

The metabolic network is the only example of a directed bipartite network appearing in this book, and indeed the only naturally occurring example of such a network the author has come across, although no doubt there are others to be discovered if one looks hard enough.

[35](#)

Also the only such network in the book.

[36](#)

Also called a *yeast two-hybrid screen* or Y2HS for short, in recognition of the fact that the technique is usually implemented inside yeast cells, as discussed later.

[37](#)

Technically, DNA is a double-stranded polymer, having two parallel chains of nucleotides forming the famous double helix shape. However, the two strands contain essentially the same sequence of nucleotides and so for our purposes the fact that there are two is not important (although it is very important in other circumstances, such as in the reproduction of a cell by cellular division and in the repair of damaged DNA).

[38](#)

“Assay” is biological jargon for an experimental test.

[39](#)

Neurons do sometimes have direct connections between them without synapses. These direct connections are called *gap junctions*, a confusing name, since it sounds like it might be a description of a synapse but is in reality quite different. In our brief treatment of neural networks, however, we will ignore gap junctions.

[40](#)

In common parlance, one refers to a *food chain*, meaning a chain of predator-prey relations between organisms starting with some lowly organism at the bottom of the chain, such as a microbe of some kind, and working all the way up to some ultimate predator at the top, such as a lion or a human being. Only a moment’s reflection, however, is enough to convince us that real ecosystems cannot be represented by single chains, and a complete network of interactions is needed in most cases.

[41](#)

In Fig. 5.8, for example, there are edges in both directions between the fish and squid vertices, which makes it impossible to draw the network with all edges running in the same direction.

[42](#)

One is reminded of the schoolhouse rhyme by Augustus de Morgan:

Great fleas have little fleas upon their backs to bite ’em,
And little fleas have lesser fleas, and so ad infinitum.

[43](#)

This use of the word “actor” sometimes leads to confusion: an actor need not be a person who actually acts, and need not even be a person. In a social network of business relationships between

companies, for instance, the actors are the companies (and the ties are the business relationships).

[44](#)

There does not seem to be a special name given to networks with self-edges. They are just called “networks with self-edges.”

[45](#)

As discussed in the next section, this is not the case for directed networks. In directed networks, self-edges are represented by a 1 in the corresponding diagonal element of the adjacency matrix.

[46](#)

The values on edges also sometimes represent lengths of some kind. On a road or airline network, for instance, edge values could represent the number of kilometers or miles the edges cover, or they could represent travel time along the edges, which can be regarded as a kind of length—one denominated in units of time rather than distance. Edge lengths are, in a sense, the inverse of edge weights, since two vertices that are strongly connected can be regarded as “close” to one another and two that are weakly connected can be regarded as far apart. Thus one could convert between weights and lengths by taking reciprocals, although this should be regarded as only an approximate procedure; in most cases there is no formal sense in which edge weights and lengths are equivalent.

[47](#)

The diagonal elements are a special case, since they are equal to 0 or 2 in an undirected network even when there are no multiedges or weighted edges.

[48](#)

Indeed, one can understand the appearance of the 2 in the undirected case as a consequence of the equivalence between undirected and directed networks mentioned above: an undirected self-edge can be thought of as two directed self-edges at the same vertex, each of which contributes 1 to the corresponding element of the adjacency matrix.

[49](#)

In the mathematical literature one often sees the abbreviation DAG, which is short for *directed acyclic graph*.

[50](#)

As discussed in Section 4.2, there are in real citation networks rare instances in which two papers both cite each other, forming a cycle of length two in the citation network, for instance if an author publishes two related papers in the same issue of a journal. Citation networks are, nonetheless, acyclic to a good approximation.

[51](#)

We could just use ordinary edges joining vertex pairs to represent our family ties, placing an edge between any two vertices that correspond to individuals in the same family. This, however, doesn’t explicitly tell us when two edges correspond to ties within the same family, and there is no single object in the network that corresponds to a family the way a hyperedge does in the hypergraph. In a number of ways, therefore, the hypergraph is more convenient.

[52](#)

In countries such as Spain or Canada, where same-sex marriages are permitted, the network would not be truly bipartite because there would be some edges between like kinds of vertex.

[53](#)

In principle, one could put directions on the edges of a tree and make it a directed network, but the definition of a tree as a loopless network ignores directions if there are any. This means that a tree is not the same thing as a directed acyclic graph (Section 6.4.2). A directed tree is always a

directed acyclic graph, but the reverse is not also true, since the definition of a loop in a directed acyclic graph takes the directions of the edges into account. A directed acyclic graph may well have loops if we ignore directions (see for example Fig. 6.3).

[54](#)

Such parts are called “components”—see Section 6.11.

[55](#)

This is a slightly odd way of drawing trees, with the root at the top and the leaves at the bottom. The more familiar trees of the wooden kind are, of course, the other way up. The upsidedown orientation has, however, become conventional in mathematics and computer science, and we here bow to this convention.

[56](#)

A plane is a flat surface with open boundaries. One can define a generalization of a planar network for other types of two-dimensional surface, such as the torus, which wraps around on itself. A standard planar network, however, does not wrap around.

[57](#)

The theorem only applies for a map on a surface with topological genus zero, such as a flat plane or a sphere. A map on a torus (which has genus 1) can require as many as seven colors.

[58](#)

Appel and Haken’s proof is an interesting one and was controversial at the time of its publication because it made extensive use of a computer to check large numbers of special cases. On the one hand, the proof was revolutionary for being the first proof of a major mathematical result generated in this fashion. On the other hand a number of people questioned whether it could really be considered a proof at all, given that it was far too large for a human being to check its correctness by hand.

[59](#)

In graph theory K_n denotes the complete graph with n vertices, i.e., the graph of n vertices with all $\binom{n}{2}$ possible single edges present. UG stands for “utility graph.” UG is the complete bipartite graph on two groups of three vertices.

[60](#)

See Kennedy *et al.* [170] for an interesting history of the theorem and references to the original papers.

[61](#)

Notice that this expression still gives the correct result if there are self-edges in the graph, provided each such edge is represented by a diagonal element $A_{ii} = 2$ as discussed earlier, and not 1.

[62](#)

Occasionally connectance is defined as $\rho = m/n^2$, which for large networks differs from Eq. (6.24) by about a factor of 2. With that definition $0 \leq \rho < \frac{1}{2}$.

[63](#)

For a rigorous proof we can use induction. If there are $N_{ik}^{(r-1)}$ paths of length $r - 1$ from i to k , then by arguments similar to those above there are $N_{ij}^{(r)} = \sum_k N_{ik}^{(r-1)} A_{kj}$ paths of length r from i to j , or in matrix notation $\mathbf{N}^{(r)} \mathbf{N}^{(r-1)} \mathbf{A}$, where $\mathbf{N}^{(r)}$ is the matrix with elements $N_{ij}^{(r)}$. This implies that if $\mathbf{N}^{(r-1)} = \mathbf{A}^{r-1}$ then $\mathbf{N}^{(r)} = \mathbf{A}^r$ and with the initial case $\mathbf{N}^{(1)} = \mathbf{A}$ we have $\mathbf{N}^{(r)} = \mathbf{A}^r$ for all r . Taking the ij th element of both sides then gives Eq. (6.31).

[64](#)

If we wish to count each loop only once, we should roughly speaking divide by r , but this does not allow for paths that have symmetries under a change of starting points, such as paths that consist of the same subloop traversed repeatedly. Counting such symmetric paths properly is a complex problem that can be solved exactly in only a few cases.

[65](#)

Such matrices have multiple or “degenerate” eigenvalues and technically have a non-zero nilpotent part in their Jordan decomposition.

[66](#)

Indeed, any mapping $\mathbf{A} \rightarrow \mathbf{Q}^{-1}\mathbf{A}\mathbf{Q}$ of a matrix preserves its eigenvalues. Such mappings are called *similarity transformations*.

[67](#)

No cheating: you’re not allowed to swim or use a boat.

[68](#)

The word “connectivity” is occasionally also used in the networks literature as a synonym for “degree.” Given that the word also has the older meaning discussed here, however, this seems an imprudent thing to do, and we avoid it in this book.

[69](#)

For a first principles proof that is not based on Menger’s theorem see, for instance, Ahuja *et al.* [8].

[70](#)

Another interesting and slightly surprising computational use of the max-flow/min-cut theorem is in the polynomial-time algorithm for finding ground states of the thermal random-field Ising model [257], an interesting cross-fertilization between network theory and physics: it is relatively common for physics ideas to find application in network theory, but the reverse has been considerably rarer.

[71](#)

In fact, the graph Laplacian matrix for undirected networks defined in this section does not have a clean generalization for directed networks, although several possible candidates have been suggested. Most of the results in the remaining sections of this chapter do not generalize easily to the directed case.

[72](#)

In fact the graph Laplacian doesn’t occupy *quite* the same position as ∇^2 does in the normal diffusion equation—there is a plus sign in Eq. (6.42) where a minus sign appears in the normal equation. We could easily get rid of this discrepancy by reversing the sign of the definition in Eq. (6.41), but the definition as given has become the standard one and so for consistency we will stick with it.

[73](#)

This is clearly the right answer from a physical point of view, since the fluid in our diffusion process is conserved—there is a fixed, finite amount of it—so it is impossible for the amount on any vertex to become infinite.

[74](#)

It is not a properly normalized eigenvector. The properly normalized vector would be $(1,1,1,...)/\sqrt{n}$.

[75](#)

Occasionally λ_2 is also called the *spectral gap*.

[76](#)

One might think that this equation could be simplified by reordering the terms so that most of them cancel out, but this is not allowed. The sum viewed as individual terms is not absolutely convergent and hence does not have a unique limit. Only the complete sum over t as written is meaningful and a reordering of the terms will give the wrong answer.

[77](#)

For those interested in traditional social network analysis, introductions can be found in the books by Scott [293] and by Wasserman and Faust [320].

[78](#)

Technically, there could be more than one eigenvector with eigenvalue κ_1 , only one of which need have all elements non-negative. It turns out, however, that this cannot happen: the adjacency matrix has only one eigenvector of eigenvalue κ_1 . See footnote 2 on page 346 for a proof.

[79](#)

This is not entirely true, as we will see in Section 7.5. Web pages that point to many others are often directories of one sort or another and can be useful as starting points for web surfing. This is a different kind of importance, however, from that highlighted by the eigenvector centrality and a different, complementary centrality measure is needed to quantify it.

[80](#)

For the left eigenvector it would be the in-component.

[81](#)

The eigenvalues being defined by $\mathbf{A}\mathbf{v} = \kappa\mathbf{v}$, we see that $(\mathbf{A} - \kappa\mathbf{I})\mathbf{v} = 0$, which has non-zero solutions for $\mathbf{v}$ only if $(\mathbf{A} - \kappa\mathbf{I})$ cannot be inverted, i.e., if $\det(\mathbf{A} - \kappa\mathbf{I}) = 0$, and hence this equation gives the eigenvalues κ .

[82](#)

Formally one recovers finite values again when one moves past $1/\kappa_1$ to higher α , but in practice these values are meaningless. The method returns good results only for $\alpha < 1/\kappa_1$.

[83](#)

It is easy to confirm that this vector is indeed an eigenvector with eigenvalue 1. That there is no eigenvalue larger than 1 is less obvious. It follows from a standard result in linear algebra, the Perron-Frobenius theorem, which states that the largest eigenvalue of a matrix such as $\mathbf{A}\mathbf{D}^{-1}$ that has all elements non-negative is unique—there is only one eigenvector with this eigenvalue—that the eigenvector also has all elements non-negative, and that it is the only eigenvector with all elements non-negative. Combining these results, it is clear that the eigenvalue 1 above must be the largest eigenvalue of the matrix $\mathbf{A}\mathbf{D}^{-1}$. For a discussion of the Perron-Frobenius theorem see Ref. [217] and the two footnotes on page 346 of this book.

[84](#)

This statement is only approximately correct since, as discussed in Section 6.4.1, the cocitation matrix is not precisely equal to the adjacency matrix of the cocitation network, having non-zero elements along its diagonal where the adjacency matrix has none.

[85](#)

Recall that geodesic paths need not be unique—vertices can be joined by several shortest paths of the same length. The length d_{ij} however is always well defined, being the length of any one of these paths.

[86](#)

www.imdb.com

[87](#)

Perhaps most famous for his role as the evil wizard Saruman in the film version of *The Lord of the Rings*.

[88](#)

Much of sociological literature concerns power or “social capital.” It may seem ruthless to think of individuals exploiting their control over other people’s information to gain the upper hand on them, but it may also be realistic. At least in situations where there is a significant pay-off to having such an upper hand (like business relationships, for example), it is reasonable to suppose that notions of power derived from network structure really do play into people’s manipulations of the world around them.

[89](#)

It is perhaps no coincidence that the highest betweenness belongs to an actor who appeared in both European and American films, played roles in several different languages, and worked extensively in both film and television, as well as on stage. Rey was the archetypal “broker,” with a career that made him a central figure in several different arms of the entertainment business that otherwise overlap relatively little.

[90](#)

This score is shared by many actors. It is the minimum possible score of $2n - 1$ as described above.

[91](#)

Another possibility, proposed by Freeman [128] in his original paper on betweenness, is to divide by the maximum possible value that betweenness can take on any network of size n , which, as mentioned above, occurs for the central vertex in a star graph. The resulting expression for between is then

$$x_i = \frac{1}{n^2 - n + 1} \sum_{st} \frac{n_{st}^i}{g_{st}}.$$

We, however, prefer Eq. (7.38), which we find easier to interpret, although the difference between the two becomes small anyway in the limit of large n .

[92](#)

All paths, that is, that terminate at the target vertex t the first time they reach it. Since we use an absorbing random walk, paths that visit the target, move away again, and then return are not included in the random-walk betweenness.

[93](#)

This definition is slightly awkward to remember, since the members of a k -plex are allowed to be unconnected to $k - 1$ other members and not k . It would perhaps have been more sensible to define k such that a 0-plex was equivalent to a normal clique, but for better or worse we are stuck with the definition we have.

[94](#)

Note that for the purposes of this latter definition, an edge between two vertices A and B within the group counts as *two* connections, one from A to B and one from B to A.

[95](#)

We have to be careful about the meaning of the word “maximal” here. It is possible to have a group of vertices such that each is connected to at least k others and no *single* vertex can be added

while retaining this property, but it may be possible to add more than one vertex. Such groups, however, are not considered to be k -cores. A group is only a k -core if it is not a subset of any larger group that is a k -core.

[96](#)

A closely related process, *bootstrap percolation*, has also been studied in statistical physics, principally on regular lattices.

[97](#)

To see this suppose we have a component that is perfectly transitive but not a clique, i.e., there is at least one pair of vertices u, w in the component that are not directly connected by an edge. Since u and w are in the same component they must therefore be connected by some path of length greater than one, $u, v_1, v_2, v_3, \dots, w$. Consider the first two links in this path. Since u is connected by an edge to v_1 and v_1 to v_2 it follows that u must be connected to v_2 if the network is perfectly transitive. Then consider the next two links. Since u is connected to v_2 and v_2 to v_3 it follows that u must be connected to v_3 . Repeating the argument all the way along the path, we can then see that u must be connected by an edge to w . But this violates the hypothesis that u and w are not directly connected. Hence no perfectly transitive components exist that are not cliques.

[98](#)

It's not entirely clear why the clustering coefficient has the name it has. The name doesn't appear to be connected with the earlier use of the word clustering in social network analysis to describe groups or clusters of vertices (see Section 11.11.2). The reader should be careful to avoid confusing these two uses of the word.

[99](#)

In fact, we could count each path just in one direction, provided we did it for both the numerator and denominator of Eq. (7.39). Doing so would decrease both counts by a factor of two, but the factors would cancel and the end result would be the same. In most cases, and particularly when writing computer programs, it is easier to count paths in both directions—it avoids having to remember which paths you have counted before.

[100](#)

The notation C_i is used for both the local clustering coefficient and the closeness centrality and we should be careful not to confuse the two.

[101](#)

As an example, in Section 11.11.1 we study methods for partitioning networks into clusters or communities and we will see that effective computer algorithms for this task can be created based on betweenness measures, but that almost equally effective and much faster algorithms can be created based on local clustering.

[102](#)

Actually, the local clustering coefficient hadn't yet been invented. It was first proposed to this author's knowledge by Watts [321] a few years later.

[103](#)

As discussed in Section 8.6.1, vertices with low degree tend to have high values of C_i in most networks and this means that C_{ws} is usually larger than the value given by Eq. (7.41), sometimes much larger.

[104](#)

This figure is an unusually high one among directed networks, but there are reasons for it. One is that many of the links between web pages are between pages on the same website, and it is

common for such pages to link to each other. If you exclude links between pages on the same site the value of the reciprocity is lower.

[105](#)

This is similar in spirit to the concept of “frustration” that arises in the physics of magnetic spin systems.

[106](#)

The proof we give is not Harary’s proof, which was quite different and not constructive.

[107](#)

As an interesting historical note, we observe that while Harary’s proof of his theorem is perfectly correct, his interpretation of it was, in this author’s opinion, erroneous. In his 1953 paper [154], he describes the meaning of the theorem in the following words: “A psychological interpretation of Theorem 1 is that a ‘balanced group’ consists of two highly cohesive cliques which dislike each other.” (Harary is using the word “clique” in a non-technical sense here to mean a closed group of people, rather than in the graph theoretical sense of Section 7.8.1.) However, just because it is possible to color the network in two colors as described above does not mean the network forms two groups. Since the vertices of a single color are not necessarily contiguous, there are in general many groups of each color, and it seems unreasonable to describe these groups as forming a single “highly cohesive clique” when in fact they have no contact at all. Moreover, it is neither possible nor correct to conclude that the members of two groups of opposite colors dislike each other unless there is at least one edge connecting the two. If two groups of opposite colors never actually have any contact then it might be that they would get along just fine if they met. It’s straightforward to prove that such an occurrence would lead to an unbalanced network, but Harary’s statement says that the *present* balanced network implies dislike, and this is untrue. Only if the network were to remain balanced upon addition of one or more edges between groups of unlike colors would his conclusion be accurate.

[108](#)

This is actually a bad name for it—it should be called *Hamming distance*, since it is essentially the same as the Hamming distance of computer science and has nothing to do with Euclid.

[109](#)

This definition is not obviously symmetric with respect to i and j but, as we see, does in fact give rise to an expression for the similarity that is symmetric.

[110](#)

It is interesting to note that when we expand this measure in powers of the adjacency matrix, as we did in Eq. (7.63), the second-order (i.e., path-length two) term is the same as the structural equivalence measure of Eq. (7.53), which perhaps lends further credence to both expressions as natural measures of similarity.

[111](#)

The study used a “name generator”—students were asked to list the names of others they considered to be their friends. This results in a directed network, but we have neglected the edge directions in the figure. In our representation there is an undirected edge between vertices i and j if either of the pair considers the other to be their friend (or both).

[112](#)

Ignoring, for the purposes of argument, dogs, cats, imaginary friends, and so forth.

[113](#)

Technically, we are making connections at random while preserving the vertex degrees. We could in principle ignore vertex degrees and make connections truly at random, but in practice this is found to give much poorer results.

[114](#)

An alternative measure of assortativity has been proposed by Gupta *et al.* [152]. That measure however gives equal weight to each group of vertices, rather than to each edge as the modularity does. With this measure if one had a million vertices of each of two types, which mixed with one another entirely randomly, and ten more vertices of a third type that connected only among themselves, one would end up with a score of about 0.5 [239], which appears to imply strong assortativity when in fact almost all of the network mixes randomly. For most purposes therefore, the measure of Eq. (7.69) gives results more in line with our intuitions.

[115](#)

Of course, one could make up some measure of national differences, based say on geographic distance, but if the question we are asked is, “Are these two people of the same nationality?” then under normal circumstances the only answers are “yes” and “no.” There is nothing in between.

[116](#)

In the US school system there are 12 grades of one year each and to begin grade g students normally must be at least of age $g + 5$. Thus the 9th grade corresponds to children of age 14 and 15.

[117](#)

There could be non-linear correlations in such a network and we could still have $r = 0$; the correlation coefficient detects only linear correlations. For instance, we could have vertices with high and low values of x_i connected predominantly to vertices with intermediate values. This is neither assortative nor disassortative by the conventional definition and would give a small value of r , but might nonetheless be of interest. Such non-linear correlations could be discovered by examining a plot such as Fig. 7.11 or by using alternative measures of correlation such as information theoretic measures. Thus it is perhaps wise not to rely solely on the value of r in investigating assortative mixing.

[118](#)

For the Internet there are no vertices of degree zero, since a vertex is not considered part of the Internet unless it is connected to at least one other.

[119](#)

We used the word hub in a different and more technical sense in Section 7.5 to describe vertices in directed networks that point to many “authorities.” Both senses are common in the networks literature, and in many cases the reader must deduce from the context which is being used. In this book we will mostly use the word in the less technical sense introduced here, of a vertex with unusually high degree. When we use it in the other sense of Section 7.5 we will say so explicitly.

[120](#)

Such plots are also sometimes called *rank/frequency* plots because one of their earliest uses was to detect power-law behavior in the frequency of occurrence of words in natural languages. If the data you are measuring are frequencies, then the cumulative distribution graph is a plot of rank against frequency. Since then such plots have been used to detect power-law behavior in many quantities other than frequencies, but the name “rank/frequency plot” is still often used.

[121](#)

In fact, this formula is only an approximation to the full formula for the exponent. The full formula, unfortunately, does not give a closed-form solution for α and is therefore hard to use. Equation (8.6) works well provided $k_{\min}$ is greater than about 6, which is true for many networks. In cases where it is not, however, the full formula must be used—see Ref. [72].

[122](#)

Using the data of Broder *et al.* [56].

[123](#)

Using the data of Redner [280].

[124](#)

For the AS-level data of Fig. 8.3.

[125](#)

It is sometimes claimed that essentially all networks show clustering higher than expected [12, 323], which is at odds with the results given here. There seem to be two reasons for the disagreement. First, the claims are based primarily on comparisons of measured clustering coefficients against values calculated on the Poisson random graph, a simple model network with a Poisson degree distribution, which we study in Chapter 12.1. Many networks, however, have right-skewed degree distributions which are very far from Poissonian, and hence the random graph is a poor model against which to compare measurements and probably gives misleading results. Second, the clustering coefficients in these comparisons are mostly calculated as an average of the local clustering, following Eq. (7.44). On networks with highly skewed degree distributions this definition can give very different results from the definition, Eq. (7.41), used in our calculations. Usually Eq. (7.44) gives much larger numbers than Eq. (7.41), which could explain the discrepancies in the findings.

[126](#)

An alternative and more complex proposal is that the behavior of the local clustering coefficient arises through hierarchical structure in a network—that not only are there groups, but that the groups are divided into smaller groups, and those into still smaller ones, and so on. See Refs. [95, 278, 309].

[127](#)

If we wish to say that the running time is exactly proportional to n , we can use the notation $\Theta(n)$.

[128](#)

There are occasional instances where this is not true, so it is worth just bearing in mind the possibility of sub-leading terms.

[129](#)

There are whole subfields in computer science devoted to the development of algorithms that run quickly even when part of the data is stored on a slow disk. Usually such algorithms work by reordering operations so that many operations can be performed on the same data, stored in the main memory, before swapping those data for others on the disk.

[130](#)

For directed networks, which are represented by asymmetric adjacency matrices, this issue does not arise—the full matrix, both the upper and lower triangles, is used to store the structure of the network.

[131](#)

Of course we could equally well store the edges in the lower triangle of the matrix and neglect the upper triangle. Either choice works fine.

[132](#)

One advantage of the adjacency matrix is that the amount of space it consumes is independent of the number of edges in the network. (It still depends on the number of vertices, of course.) As we will see in Section 9.4, other data formats such as the adjacency list use varying amounts of memory, even for networks with the same number of vertices, depending on how many edges

there are. In calculations where edges are frequently added or removed it may be convenient—and increase the speed of our algorithms—to have the size of our data structures remain constant, although this advantage must be weighed against the substantial space savings of using the adjacency list or other memory-efficient formats.

[133](#)

Note that the number of entries in the list of neighbors for a vertex varies from one vertex to another, and may even be zero, being equal to the degree of the corresponding vertex. Most modern computer languages, including C and its derivatives and JAVA, allow the creation of two-dimensional matrices with rows having varying numbers of elements in this fashion. Some older languages, like FORTRAN 77, do not allow this, making it more difficult to store adjacency lists in a memory-efficient way.

[134](#)

Note that the amount of memory used is now a function of m rather than n . For algorithms in which edges are added or removed from a network during the course of a calculation this means that the size of the adjacency list can change, which can complicate the programming and potentially slow down the calculation. Normally, however, this added complication is not enough to outweigh the considerable benefits of the adjacency list format.

[135](#)

Indeed, the adjacency list for an undirected network such as that given above could be viewed as a special case of the directed adjacency list for a network in which each undirected edge is replaced by two directed ones, one in each direction. It takes only a moment to convince oneself that this results precisely in the sort of double representation of each edge that we saw in the undirected case.

[136](#)

The answers are essentially the same in the directed case. The demonstration is left as an exercise.

[137](#)

We are thus calculating a sort of “average worst-case” behavior, allowing for the worst case in which we have to look through the entire list, but then averaging that worst case over many different lists. This is a reasonable (and standard) approach because almost all of the algorithms we will be considering do many successive “find” operations during a single run, but it does mean that we are technically not computing the complexity of the absolute worst case situation.

[138](#)

This is not a standard name. As far as the present author is aware, this data structure doesn’t have a standard name, since it is not used very often. Moreover, the name isn’t even entirely accurate. The structure is technically not a tree but a forest, i.e., a collection of many trees. Still, “adjacency tree” is simple and descriptive, and analogous to “adjacency list,” which is also, technically, not a single list but a collection of lists.

[139](#)

The word “tree” has a different meaning here from the one it had in Section 6.7, although the two are related. There a tree meant a network with no loops in it. Here it refers to a data structure, although as we will see, the tree data structure can also be regarded as a network with no loops. In effect we are using a network to store information about another network, which is a nice touch.

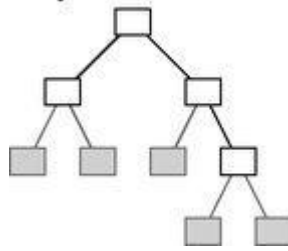
[140](#)

As pointed out in a previous footnote on page 127, it is slightly odd to put the “root” the top of the tree. Most of us are more familiar with trees that have their roots at the bottom. We could of course draw the tree the other way up—it would have the same meaning—but it has become conventional to draw the root at the top of the picture, and we bow to that convention here.

A “pointer” in computer programming is a special variable that holds the address in memory of another variable.

The proof is as follows.

Consider the set of “empty nodes,” meaning the missing children immediately below current nodes in the tree (gray boxes in the figure on the right). Suppose that when there are k nodes in total in the tree there are c_k such empty nodes and that their average depth, measured from the root, is d_k . When we add one new value to the tree at random it will occupy one of these empty nodes thereby decreasing their number by one. At the same time two new empty nodes will appear, the children of the newly added node. Overall therefore $c_{k+1} = c_k + 1$. Noting that $c_1 = 2$, this immediately implies that $c_k = k + 1$.



At the same time the sum of the lengths of all paths from the root to an empty node, which by definition is equal to $c_k d_k$, decreases (on average) by d_k when a new random node is added and increases by $2(d_k + 1)$ for the two new ones, so that $c_{k+1} d_{k+1} = c_k d_k + d_k + 2$. Eliminating c_k , we then find that $d_{k+1} = d_k + 2/(k + 2)$. Noting that $d_1 = 2$, this implies

$$d_k = 1 + \sum_{m=0}^{k-1} \frac{2}{m+2} = -1 + 2 \sum_{m=1}^{k+1} \frac{1}{m} = 2\gamma - 1 + 2\ln k + O(1/k),$$

where $\gamma = 0.5772 \dots$ is Euler’s constant.

Thus the average depth of the empty nodes is $O(\log k)$. Since the find and addition operations on the tree both involve searching the tree until an empty node is encountered, it immediately follows that both operations have average complexity $O(\log k)$ on our randomized tree, just as they do on the optimally packed tree.

Note that this is only a statement about the average behavior of the tree and not about the worst-case behavior. If we are unlucky the depth of the tree could be much larger than the average, up to the maximum depth k . The randomized tree only provides a guarantee of average performance, meaning performance averaged over many possible runs of an algorithm. Individual runs will vary around the mean.

It does place some overhead on the edge addition and deletion operations, meaning the complexity is still $O(1)$ but the operations take a constant factor longer to complete, since we have to update both adjacency matrix and list, where normally we would only have to update one or the other. Whether this makes an significant difference to the running of a program will depend on the particular algorithm under consideration.

Or an adjacency tree—see Section 9.5.

[145](#)

For integers, such as vertex degrees, it is under certain conditions possible to sort faster, in time $O(n)$, using the so-called *radix sort* algorithm. See, for example, Cormen *et al.* [81].

[146](#)

An equivalent to the clustering coefficient can be defined for a directed network (see Section 7.9) but we limit ourselves here to the much commoner undirected case.

[147](#)

Note that this calculation automatically accounts for the factor of three appearing in the numerator of Eq. (10.3), since each triangle is counted three times, once each from the point of view of the three vertices it connects.

[148](#)

Other possible ways to improve the algorithm are to use the adjacency tree structure of Section 9.5, or to use the adjacency list but always test for the presence of an edge between two vertices by searching the neighbors of the lower-degree vertex to see if the higher is among them (rather than the algorithm described above, which chooses which one to search at random).

[149](#)

In physics, breadth-first search is sometimes called the “burning algorithm.”

[150](#)

In many cases we may find the result we want before calculating all distances, in which case we can save ourselves the effort of calculating the rest, but in the worst case we will have to calculate them all. See Section 10.3.4 for more discussion.

[151](#)

On the other hand, for a dense network where $m \propto n^2$, we have a running time of $O(n^2)$.

[152](#)

Note that this actually counts each path twice (since the path between i and j is counted once when i is considered the source vertex and once when j is), except for the path from each vertex to itself, which is counted only once (when that vertex is the source). This, however, is correct: the betweenness centrality, as defined in Eq. (7.36), indeed counts each path twice, except for the path from a vertex to itself. As discussed in Section 7.7, some researchers define betweenness differently, counting paths only once, but that merely reduces all values by a factor of two.

[153](#)

In this case we are considering the first and last vertices on a path to be members of that path. As discussed in Section 7.7, the first and last vertices are sometimes excluded from the calculation, which means that the betweenness score of each vertex is smaller by an additive constant equal to twice the number of vertices in the component. If we wish to calculate betweenness according to this alternative definition, the simplest approach is to use the algorithm described here and then subtract the additive constant from each vertex’s score at the end.

[154](#)

As discussed in footnote 9, these scores give the betweenness as defined in Eq. (7.36). To get true path counts one would have to divide by two and add a half (or equivalently add one then divide by two) to correct for the double counting of paths between distinct vertices.

[155](#)

We assume that the lengths are all non-negative. If lengths can be negative, which happens in some cases, then the problem is much harder, falling in the class of “NP-complete” computational problems, for which even the best known algorithms take an amount of time exponential in n to

finish, in the worst case [8]. Indeed, if edges are allowed to have negative lengths, there may not be any shortest path between a pair of vertices, since one can have a loop in the network that has negative length, so that one can reduce the length of a path arbitrarily by going around the loop repeatedly.

[156](#)

In theory one can achieve a slightly better running time of $O(m + n \log n)$ using a data structure known as a Fibonacci heap [81], but in practice the operation of the Fibonacci heap is so complicated that the calculation usually ends up running slower.

[157](#)

The augmenting path algorithm is not the only algorithm for calculating maximum flows. It is, however, the simplest and its average performance is about as good as any other, so it is a good choice for everyday calculations. It's worth noting, however, that the *worst-case* performance of the algorithm is quite poor: for pathological networks, the algorithm can take a very long time to run. Another algorithm, the *preflow-push algorithm* [8], has much better worst-case performance and comparable average-case performance, but is considerably more complicated to implement.

[158](#)

See Ahuja *et al.* [8] or Cormen *et al.* [81] for details of the general case.

[159](#)

Technically, the augmenting path algorithm doesn't specify how paths are to be found. Here we study the particular version in which paths are found using breadth-first search, which is known to be one of the better-performing variants. Sometimes this variant is called the *shortest augmenting path algorithm* or the *Edmonds-Karp algorithm*.

[160](#)

On networks with directed edges, we allow either the same flow in both directions along an edge (i.e., zero net flow) or one more unit in the forward direction than in the backward direction, but not vice versa.

[161](#)

Note, however, that the independent paths are not necessarily unique: there can be more than one choice of paths and some of them may not be found by this algorithm. Furthermore, there can be points in the network where paths come together at a vertex and then part ways again. If such points exist, you will have to make a choice about which way to go at the parting point. It doesn't matter what choice you make in the sense that all choices lead to a correct set of paths, but different choices will give different sets of paths.

[162](#)

Technically the power method finds the eigenvector corresponding to the eigenvalue of largest absolute magnitude and hence the method would fail to find the eigenvector we want if the largest absolute magnitude belongs to a negative eigenvalue. For a matrix with all elements non-negative, however, such as the adjacency matrix, it turns out this can never happen. Here is a proof of this result for an undirected network where $\mathbf{A}$ is symmetric; the general case is covered, for example, in Ref. [217]. Let μ be the most negative eigenvalue of a real symmetric matrix $\mathbf{A}$ and let $\mathbf{w}$ be the corresponding eigenvector, with elements w_i . Then, given that $\mathbf{w}^T \mathbf{w} = \sum_i w_i^2 > 0$,

$$|\mu| \mathbf{w}^T \mathbf{w} = |\mu \mathbf{w}^T \mathbf{w}| = |\mathbf{w}^T \mathbf{A} \mathbf{w}| = \left| \sum_{ij} A_{ij} w_i w_j \right| \leq \sum_{ij} |A_{ij} w_i w_j| = \sum_{ij} A_{ij} |w_i| |w_j| = \mathbf{x}^T \mathbf{A} \mathbf{x},$$

where $\mathbf{x}$ is the vector with components $|w_i|$. The inequality here follows from the so-called triangle inequality $|a + b| \leq |a| + |b|$, which is true for all real numbers a, b . Rearranging, we now find that

$$|\mu| \leq \frac{\mathbf{x}^T \mathbf{A} \mathbf{x}}{\mathbf{w}^T \mathbf{w}} = \frac{\mathbf{x}^T \mathbf{A} \mathbf{x}}{\mathbf{x}^T \mathbf{x}},$$

where we have made use of $\mathbf{x}^T \mathbf{x} = \sum_i |w_i|^2 = \mathbf{w}^T \mathbf{w}$. Now we write $\mathbf{x}$ as a linear combination of the normalized eigenvectors $\mathbf{v}_i$ of $\mathbf{A}$ thus: $\mathbf{x} = \sum_i c_i \mathbf{v}_i$, where the c_i are real coefficients whose exact values are not important for this proof. Then, if κ_i is the eigenvalue corresponding to $\mathbf{v}_i$ and κ_1 is the most positive eigenvalue, we have

$$\frac{\mathbf{x}^T \mathbf{A} \mathbf{x}}{\mathbf{x}^T \mathbf{x}} = \frac{\sum_j c_j \mathbf{v}_j^T \mathbf{A} \sum_i c_i \mathbf{v}_i}{\sum_j c_j \mathbf{v}_j^T \sum_i c_i \mathbf{v}_i} = \frac{\sum_j c_j \mathbf{v}_j^T \sum_i c_i \kappa_i \mathbf{v}_i}{\sum_j c_j \mathbf{v}_j^T \sum_i c_i \mathbf{v}_i} = \frac{\sum_i c_i^2 \kappa_i}{\sum_i c_i^2} \leq \frac{\sum_i c_i^2 \kappa_1}{\sum_i c_i^2} = \kappa_1,$$

where we have made use of the orthogonality property $\mathbf{v}_j^T \mathbf{v}_i = \delta_{ij}$. (The inequality is an exact equality if and only if $\mathbf{x}$ is an eigenvector with eigenvalue κ_1 .) Putting these results together, we find that $|\mu| \leq \kappa_1$ and hence the most negative eigenvalue never has a magnitude greater than that of the most positive eigenvalue (although if we are unlucky the two magnitudes could be equal). The result proved here is one part of the *Perron-Frobenius theorem*. The other part, that the leading eigenvector has all elements non-negative, is proved in the following footnote.

[163](#)

This result, like that in footnote 1, is a part of the Perron-Frobenius theorem. To prove it—at least for the case of symmetric $\mathbf{A}$ —let κ_1 be the most positive eigenvalue of $\mathbf{A}$ and let $\mathbf{v}$ be a corresponding eigenvector. (We will allow, for the moment, the possibility that there is more than one eigenvector with eigenvalue κ_1 , though we show below that in fact this cannot happen in a connected network.) Note that $\kappa_1 \geq 0$ since the sum of the eigenvalues of $\mathbf{A}$ is given by $\text{Tr } \mathbf{A} \geq 0$, and hence at least one eigenvalue must be non-negative. Then, given that $\mathbf{v}^T \mathbf{v} = \sum_i v_i^2 > 0$ and all elements of $\mathbf{A}$ are non-negative, we have

$$\kappa_1 \mathbf{v}^T \mathbf{v} = |\kappa_1 \mathbf{v}^T \mathbf{v}| = |\mathbf{v}^T \mathbf{A} \mathbf{v}| = \left| \sum_{ij} A_{ij} v_i v_j \right| \leq \sum_{ij} |A_{ij} v_i v_j| = \sum_{ij} A_{ij} |v_i| |v_j| = \mathbf{x}^T \mathbf{A} \mathbf{x},$$

where $\mathbf{x}$ is the vector with elements $|v_i|$. Rearranging this result, we find

$$\kappa_1 \leq \frac{\mathbf{x}^T \mathbf{A} \mathbf{x}}{\mathbf{v}^T \mathbf{v}} = \frac{\mathbf{x}^T \mathbf{A} \mathbf{x}}{\mathbf{x}^T \mathbf{x}},$$

where we have made use of $\mathbf{x}^T \mathbf{x} = \sum_i |v_i|^2 = \mathbf{v}^T \mathbf{v}$. As demonstrated in footnote 1 on page 346, for any vector $\mathbf{x}$ we have

$$\frac{\mathbf{x}^T \mathbf{A} \mathbf{x}}{\mathbf{x}^T \mathbf{x}} \leq \kappa_1,$$

with the equality being achieved only when $\mathbf{x}$ is an eigenvector corresponding to eigenvalue κ_1 . The only way to reconcile the two inequalities above is if they are in fact equalities in this case, implying that $\mathbf{x}$ must indeed be an eigenvector with eigenvalue κ_1 . But $\mathbf{x}$ has all elements non-negative, and hence there exists an eigenvector with eigenvalue κ_1 and all elements non-negative.

It is still possible that there might be more than one eigenvector with eigenvalue κ_1 , and that one

of the others might have negative elements. This, however, we can rule out as follows. Recall that eigenvectors with same eigenvalue can always be chosen orthogonal, and any eigenvector $\mathbf{v}$ that is orthogonal to the eigenvector with all elements non-negative would have to have both positive and negative elements in order that the product of the two vectors equal zero. Thus there is only one eigenvector with all elements non-negative.

Then, for eigenvector $\mathbf{v}$, by the results above, the vector $\mathbf{x}$ with elements $|v_i|$ is necessarily equal to the unique eigenvector with all elements non-negative. Thus if v_i is one of the positive elements of $\mathbf{v}$ then $v_i = x_i$ and

$$\sum_j A_{ij}|v_j| = \sum_j A_{ij}x_j = \kappa_1 x_i = \kappa_1 v_i = \sum_j A_{ij}v_j,$$

or, equivalently, $\sum_j A_{ij}(|v_j| - v_j) = 0$. But $|v_j| - v_j \geq 0$ so this last result can only be true if for all j we have either $A_{ij} = 0$ or $v_j - |v_j| = 0$, meaning that $v_j = |v_j| \geq 0$. Thus if $v_i > 0$ then $v_j > 0$ whenever $A_{ij} \neq 0$. In network terms, if $v_i > 0$ then $v_j > 0$ for every neighbor of i . But then we can start at i and work outwards, moving from neighbor to neighbor and so demonstrate that $v_j > 0$ for every vertex and hence $\mathbf{v} = \mathbf{x}$ and the leading eigenvector is unique.

The only exception to this last result is when the network has more than component, so that some vertices are not reachable from an initial vertex i . In that case, it is possible for the elements of the leading eigenvector corresponding to vertices in different components to have different signs. This, however, causes no problems for any of the results presented here.

[164](#)

In fact, this estimate usually errs on the pessimistic side, since the spacing of the highest eigenvalues tends to be wider than the mean spacing, so that in practice the algorithm may be faster than the estimate would suggest.

[165](#)

If we wish to be more sophisticated, we can note that it is sufficient to shift the eigenvalues by any amount greater than or equal to λ_n . Anderson and Morley [18] have shown that $\lambda_n \leq 2k_{\max}$ where $k_{\max}$ is the largest degree in the network, which we can find in time $O(n)$, considerably faster than we can find λ_n itself. Thus a quicker way to find the smallest eigenvalue would be to find the largest eigenvalue of $2k_{\max}\mathbf{I} - \mathbf{L}$.

[166](#)

Community detection is sometimes also called “clustering,” although we largely avoid this term to prevent confusion with the other, and quite different, use of the word clustering introduced in Section 7.9.

[167](#)

The problem is somewhat similar to the minimum cut problem of Section 6.12, but we are now searching for the minimum cut over all possible bisections of a network, rather than just between a given pair of vertices.

[168](#)

Technically, this statement has not actually been proved. Its truth hinges on the assumption that two fundamental classes of computational problem, called P and NP, are not the same. Although this assumption is universally believed to be true—the world would pretty much fall apart if it weren’t—no one has yet proved it, nor even has any idea about where to start. Readers interested in the fascinating branch of theoretical computer science that deals with problems of this kind are encouraged to look, for example, at the book by Moore and Mertens [227].

[169](#)

Some readers may be familiar with Kernighan's name. He was one of the authors of the original book describing the C programming language [172]. "Kernighan" is pronounced "Kernihan"—the "g" is silent.

[170](#)

One might imagine that an equivalent procedure would be to go on swapping vertex pairs until no swap can be found that decreases the cut size. This, however, turns out to be wrong. It is perfectly possible for the cut size to decrease for a few steps of the algorithm, then increase, then decrease again. If we halt the algorithm the first time we see the cut size increasing, we run the risk of missing a later state with smaller cut size. Thus the correct algorithm is the one described here, with two separate processes, one of vertex swapping, and one of checking the states so generated to see which is optimal.

[171](#)

If the network is stored in adjacency matrix form then the total run time can be improved further to $O(n^2)$, although for the common case of a sparse network this makes relatively little difference, and the adjacency matrix is costly in terms of memory space.

[172](#)

Note, however, that the vertex moving algorithm takes time $O(n^2)$ for each round of the algorithm, but we have not calculated, and do not in fact know, how many rounds are needed in general. As with the Kernighan-Lin algorithm, it is reasonable to suppose that the number of rounds needed might increase, at least slowly, with network size, which would make the time complexity of the vertex moving algorithm poorer than that of the spectral algorithm.

[173](#)

The word "clustering" as used here just refers to community detection. We have mostly stayed away from using this word in this chapter, to avoid confusion with the other use of the word clustering introduced in Section 7.9 (see footnote 5 on page 354), but the name "hierarchical clustering" is a well established and traditional one, and we use it here in deference to convention.

[174](#)

The heap must be modified slightly from the one described in Section 9.7. First, the partial ordering must be inverted so that the largest, not the smallest, element of the heap is at its root. Second, we need to be able to remove arbitrary items from the heap, not just the root item, which we do by deleting the relevant item and then moving the last item in the heap to fill the vacated space. Then we have to sift the moved item both up *and* down the heap, since it might be either too large or too small for the position in which it finds itself.

[175](#)

For the special case of single-linkage clustering, there is a slightly faster way to implement the algorithm that makes use of a so-called union/find technique and runs in time $O(n^2)$. In practice the performance difference is not very large but the union/find method is considerably simpler to program. It is perhaps for this reason that single-linkage is more often used than complete-or average-linkage clustering.

[176](#)

It would in theory be perfectly possible, however, to create a variant of the model with multiedges or self-edges, or both.

[177](#)

We need to be a little careful here: u here should really be the probability that j is not connected to the giant component via any of its connections other than the connection to i . However, it turns out that in the limit of large system size this probability is just equal to u . For large n the probability of not being connected to the giant component via any of the $n-2$ vertices other than i is not

significantly smaller than the probability for all $n-1$ vertices.

[178](#)

One can write a closed-form solution in terms of the *Lambert W -function*, which is defined as the solution to the equation $W(z)e^{W(z)} = z$. In terms of this function the size of the giant component is

$$S = 1 + \frac{W(-ce^{-c})}{c},$$

where we take the principal branch of the W -function. This expression may have some utility for numerical calculations and series expansions, but it is not widely used. Alternatively, although we cannot write a simple solution for S as a function of c , we can write a solution for c as a function of S . Rearranging Eq. (12.15) for c gives

$$c = -\frac{\ln(1-S)}{S},$$

which can be useful, for instance, for plotting purposes. (We can make a plot of S as a function of c by first making a plot of c as a function of S and then swapping the axes.)

[179](#)

In the statistical physics literature, this trick of removing a vertex is called a *cavity method*. Cavity methods are used widely in the solution of all kinds of physics problems and are a powerful method for many calculations on lattices and in low-dimensional spaces as well as on networks [218].

[180](#)

The formula derived here is not the *most* general form of the Lagrange inversion formula. It is adequate for the particular problem we are interested in solving, but the full Lagrange inversion formula is even more powerful, and can solve a broader range of problems. For details, see Wilf [329].

[181](#)

The situation gets complicated if $z(f)$ is many-valued for some f , i.e., if $f(z)$ is nonmonotonic. In our case, however, where the coefficients in the expansion of $f(z)$ are necessarily all non-negative because they are probabilities, $f(z)$ is monotonically increasing and no such problems arise.

[182](#)

There are still some holes in our argument. In particular, we have assumed that the product of the numbers of vertices on the surface of our two neighborhoods is c^{s+t} when in practice this is only the average value and there will in general be some variation. Also the calculation should really be confined to the giant component, since the longest path always falls in the giant component in the limit of large n . For a careful treatment of these issues see, for instance, Fernholz and Ramachandran [114].

[183](#)

This appears to be a reasonable figure. Bernard *et al.* [36] estimated the typical number of acquaintances for people in the United States to be about 2000—see Section 3.2.1.

[184](#)

This disagreement, highlighted particularly by Watts and Strogatz [323], was one of the observations that prompted the current wave of interest in the properties of networks in the mathematical sciences, starting in the late 1990s.

[185](#)

Professor Wilf has generously made his book available for free in electronic form. You can download it from www.math.upenn.edu/~wilf/DownldGF.html.

[186](#)

No, really. I'm not making this up.

[187](#)

Even for finite-sized networks the difference between the properties of a configuration model network and a similar network without self-edges and multiedges would only result in a correction of order $1/n$ into our results. For the large networks that are the focus of most modern network studies this means that the error introduced by allowing self-edges and multiedges is small.

[188](#)

For networks with power-law degree distributions $\square k^2 \square$ diverges, as described in Section 8.4.2, and in that case the density of multiedges may not vanish or may do so more slowly than $1/n$.

[189](#)

As before, p_{ij} should really be regarded as the expected number of edges between i and j rather than the probability and in fact the proper formulation of the model is that we place a Poisson-distributed number of edges with mean p_{ij} between each pair of vertices i, j . Thus the model can in principle have multiedges as well as self-edges, just as in the configuration model. In the limit of large m and constant c_p , however, the probability and the expected number again become equal, and the density of multiedges tends to zero, so the distinction is unimportant.

[190](#)

Another way of putting this is that the average value $\square A_{ij} \square$ of an element of the adjacency matrix is simply $\square A_{ij} \square = c_i c_j / 2m$ for all i, j —recall that the diagonal element A_{ii} of the adjacency matrix is defined to be twice the number of self-edges at vertex i , and this compensates for the extra factor of two in Eq. (13.38).

[191](#)

The probabilities of edges between vertex i and each other vertex are independent, which immediately implies that the degree has a Poisson distribution. This may be obvious to you—if you're a statistician, for example—but if not, here is a proof, which makes use of generating functions.

The probability that there are edges connecting vertex i to any specific set of vertices, including itself, is given by a product of factors p_{ij} for each edge present and $(1 - p_{ij})$ for each edge not present. This product can conveniently be written in the form

$$p_{ii}^{A_{ii}/2} (1 - p_{ii})^{1-A_{ii}/2} \prod_{j(j \neq i)} p_{ij}^{A_{ij}} (1 - p_{ij})^{1-A_{ij}},$$

where A_{ij} is the standard adjacency matrix and we adopt the convention that $0^0 = 1$ for any cases where $p_{ij} = 0$. Note that it is important to separate out the term for p_{ii} as shown, since it takes a slightly different form from the others. Recall that a self-edge is represented by a diagonal element $A_{ii} = 2$ in the adjacency matrix (see Section 6.2) and we must allow for this with the factors of two above.

The probability $p_k^{(i)}$ that vertex i has degree exactly k is the sum of these probabilities over all cases where the i th row of the adjacency matrix adds up to k (including the 2s that appear for self-edges, since a self-edge contributes +2 to the degree). We can write this sum as

$$p_k^{(i)} = \sum_{A_{i1}=0,1} \dots \sum_{A_{i2}=0,2} \dots \sum_{A_{in}=0,1} \delta(k, \sum_j A_{ij}) p_{ii}^{A_{ii}/2} (1-p_{ii})^{1-A_{ii}/2} \prod_{j(i \neq i)} p_{ij}^{A_{ij}} (1-p_{ij})^{1-A_{ij}},$$

where $\delta(a, b)$ is the Kronecker delta. It is tricky to evaluate this sum directly because of the constraint imposed by the delta function, but we can do it using a generating function. Multiplying both sides of the equation by z^k , summing over all k , and defining the generating function $g_i(z) = \sum_k p_k^{(i)} z^k$, we get

$$\begin{aligned} g_i(z) &= \sum_{k=0}^{\infty} z^k \sum_{A_{i1}=0,1} \dots \sum_{A_{i2}=0,2} \dots \sum_{A_{in}=0,1} \delta(k, \sum_j A_{ij}) p_{ii}^{A_{ii}/2} (1-p_{ii})^{1-A_{ii}/2} \prod_{j(i \neq i)} p_{ij}^{A_{ij}} (1-p_{ij})^{1-A_{ij}} \\ &= \sum_{A_{i1}=0,1} \dots \sum_{A_{i2}=0,2} \dots \sum_{A_{in}=0,1} z^{\sum_j A_{ij}} p_{ii}^{A_{ii}/2} (1-p_{ii})^{1-A_{ii}/2} \prod_{j(i \neq i)} p_{ij}^{A_{ij}} (1-p_{ij})^{1-A_{ij}} \\ &= \sum_{A_{i1}=0,1} \dots \sum_{A_{i2}=0,2} \dots \sum_{A_{in}=0,1} (p_{ii} z^2)^{A_{ii}/2} (1-p_{ii})^{1-A_{ii}/2} \prod_{j(i \neq i)} (p_{ij} z)^{A_{ij}} (1-p_{ij})^{1-A_{ij}} \\ &= (1-p_{ii} + p_{ii} z^2) \prod_{j(i \neq i)} (1-p_{ij} + p_{ij} z) \\ &= \left[1 + \frac{c_i^2}{4m} (z^2 - 1) \right] \prod_{j(i \neq i)} \left[1 + \frac{c_i c_j}{2m} (z - 1) \right]. \end{aligned}$$

Taking logs of both sides and going to the limit of large size, where $m \rightarrow \infty$ (with the c_i remaining finite), we then get

$$\begin{aligned} \ln g_i(z) &= \lim_{m \rightarrow \infty} \left\{ \ln \left[1 + \frac{c_i^2}{4m} (z^2 - 1) \right] + \sum_{j(i \neq i)} \ln \left[1 + \frac{c_i c_j}{2m} (z - 1) \right] \right\} \\ &= \frac{c_i^2}{4m} (z^2 - 1) + \sum_{j(i \neq i)} \frac{c_i c_j}{2m} (z - 1) \\ &= \frac{c_i^2}{4m} (z^2 - 1) - \frac{c_i^2}{2m} (z - 1) + \sum_{j=1}^n \frac{c_i c_j}{2m} (z - 1) \\ &= \frac{c_i^2}{4m} (z^2 - 1) - \frac{c_i^2}{2m} (z - 1) + c_i (z - 1) \\ &= c_i (z - 1) \left[1 + \frac{c_i (z - 1)}{4m} \right], \end{aligned}$$

where we have made use of Eq. (13.39) in the second-to-last line. For large m , the second term in the square brackets becomes negligible compared to the first and, taking exponentials again,

$$g_i(z) = e^{c_i(z-1)}.$$

Now we can derive the probability distribution of the degree of vertex i by differentiating:

$$p_k^{(i)} = \frac{1}{k!} \frac{d^k g_i}{dz^k} \Big|_{z=0} = e^{-c_i} \frac{c_i^k}{k!},$$

which is indeed a Poisson distribution, with mean c_i , as promised.

192

It is easy to see that there are some degree distributions that the model cannot reproduce at all—any distribution for which p_k is exactly zero for any k , for instance, since there is always a non-zero

probability that any vertex can have any degree.

[193](#)

On the other hand, if we pick a *random* edge in a network and follow it to one of its ends, then the degree of the vertex we reach is distributed according to (13.42), regardless of whether degrees are correlated or not.

[194](#)

The ratio $\langle k^2 \rangle / \langle k \rangle$ that appears in Eq. (13.43) crops up repeatedly in the study of networks. It appeared previously in Section 13.2.1 and it will come up in many later calculations.

[195](#)

There is no reason in principle why the configuration model should always overestimate the average degree of a neighbor. In some cases it could underestimate too.

[196](#)

In fact, the general derivative in this case can be expressed in terms of the so-called Bell numbers. No closed-form solution exists for third-nearest neighbors or higher, however, nor for most other choices of degree distribution.

[197](#)

This expression has an interesting history. In the 1940s Flory [123] considered a model of branching polymers in which elemental units with a fixed number of “legs”—vertices with uniform degree, in effect—joined together to form connected clumps. He showed that, if the system was restricted to forming only trees, then there was a transition at which the polymer “gelled” to create a clump of joined units which corresponds to our giant cluster and found the size of the gel. In effect, Flory’s results were a special case of the solution given here for the uniform degree distribution, although they were not expressed in the language of networks. It was not until much later that Molloy and Reed, who were, as far as I know, unaware of Flory’s work, gave the full solution for general degree distribution.

[198](#)

The only exception is when $\langle k \rangle = 2$ at the transition point.

[199](#)

Traditionally $\zeta(x)$ is actually defined to have finite values below $x = 1$ by analytic continuation. But in our case we are really interested in the value of the sum $\sum_{k=1}^{\infty} k^{-x}$, which diverges for all $x \leq 1$.

[200](#)

The numerical solution is simple: we just choose a suitable starting value ($u = \frac{1}{2}$ works fine) and iterate Eq. (13.141) until it converges. Fifty iterations are easily enough to give a highly accurate result.

[201](#)

Given the joint degree distribution we can still, if we wish, calculate the distributions of in- or out-degrees alone. They are given by

$$p_j^{(\text{in})} = \sum_{k=0}^{\infty} p_{jk}, \quad p_k^{(\text{out})} = \sum_{j=0}^{\infty} p_{jk}.$$

[202](#)

Strictly we have not shown that a giant component actually appears if the condition (13.161) is satisfied. We have only shown that (13.161) is a necessary condition for a giant component to

appear. However, we can go through an argument similar to the one we made for the undirected case in Section 13.6 to convince ourselves that there must be a giant component when the condition is satisfied.

[203](#)

For a discussion see, for instance, Refs. [222] and [244].

[204](#)

Simon himself called the mechanism the *Yule process*, in recognition of the statistician Udny Yule, who had studied a simple version many years earlier [333].

[205](#)

The main condition on the distribution is that it should have finite variance. This rules out, for example, cases in which bibliographies have a power-law distribution of sizes with exponent less than three. Empirical evidence suggests that real bibliographies have an unexceptionable distribution of sizes with a modest and finite variance, so the assumptions of Price's model are met.

[206](#)

The two fractions are in fact $a/(c + a)$ and $c/(c + a)$, respectively, as shown in Section 14.1.1.

[207](#)

There is nothing in the definition of Price's model to prevent a paper from listing the same other paper twice in its bibliography, something that doesn't happen in real citation networks. In the language of graph theory, such double citations would correspond to directed multiedges in the citation network (see Section 6.1) while true citation networks are simple graphs having no multiedges. However, as with random graph models, the probability of generating a multiedge vanishes in the limit of large network size and so the predictions of the model in this limit are not altered by allowing them, and doing so makes the mathematical treatment of the model much simpler.

[208](#)

In theory, it also increases by one if a vertex of in-degree $q - 2$ receives two new citations, and similarly for larger numbers of citations. This, however, would create a multiedge, and multiedges, as we have said, are vanishingly improbable in the limit of large network size, so we can ignore this possibility.

[209](#)

Strictly we should first prove that the degree distribution *has* an asymptotic form in the limit of large n and doesn't go on changing forever, but for the purposes of the present discussion let us assume that there is an asymptotic form.

[210](#)

The proof of this result is simple, making use of integration by parts:

$$\Gamma(x+1) = \int_0^\infty t^x e^{-t} dt = -[t^x e^{-t}]_0^\infty + x \int_0^\infty t^{x-1} e^{-t} dt = x\Gamma(x),$$

where the boundary term [...] disappears at both limits.

[211](#)

The integral form can be derived by making use of the definition of $B(x,y)$ in terms of gamma functions and the integral form of the gamma function, Eq. (14.15):

$$B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} = \frac{1}{\Gamma(x+y)} \int_0^\infty s^{x-1} e^{-s} ds \int_0^\infty t^{y-1} e^{-t} dt.$$

We change variables to $u = s/(s+t)$, $v = s+t$ (which implies $s = uv$, $t = (1-u)v$ and a Jacobian of v), giving

$$\begin{aligned} B(x, y) &= \frac{1}{\Gamma(x+y)} \int_0^1 du \int_0^\infty v dv (uv)^{x-1} e^{-uv} [(1-u)v]^{y-1} e^{-(1-u)v} \\ &= \frac{1}{\Gamma(x+y)} \int_0^\infty v^{x+y-1} e^{-v} dv \int_0^1 u^{x-1} (1-u)^{y-1} du. \end{aligned}$$

The first integral, however, is equal to $\Gamma(x+y)$ by Eq. (14.15) and hence we recover Eq. (14.33).

[212](#)

See for instance [99] or [189].

[213](#)

It may not be immediately obvious that $\pi_0(1)$ must equal 1. There is one vertex in the time interval between $\tau = 1$ and $\tau = 1 - 1/n$ and it always has in-degree zero. One vertex is a fraction $1/n$ of the whole network, so there is $1/n$ of the network in an interval of width $1/n$, which corresponds to a density $\pi_0(1) = 1$.

[214](#)

To prove this we set $x = 1$ in Eq. (14.17) to get $\Gamma(n+1)/\Gamma(1) = n(n-1) \dots 1 = n!$ and from Eq. (14.15) we have $\Gamma(1) = \int_0^\infty e^{-t} dt = 1$.

[215](#)

Hint: you will probably need Eq. (14.33).

[216](#)

Actually, the peak only exists for small values of τ and disappears once τ becomes large enough. There are no peaks in the degree distribution for the $\tau = 0.5$ and $\tau = 0.9$ curves in Fig. 14.3b.

[217](#)

Salganik *et al.* did find a weak effect of song quality—songs that had proved popular when the download numbers were reported faithfully continued to do better than expected even when the download numbers were misreported.

[218](#)

In fact, Moore *et al.* give a solution for a model in which vertices rather than edges are deleted, but the two can be treated by virtually the same means. The calculation given here is adapted from their work with only minor changes.

[219](#)

Also called the trapezium rule in British English.

[220](#)

Explicit expressions are known for the correction terms (the terms in $f'(a)$ and $f'(b)$)—they are given in terms of the Bernoulli numbers by the so-called Euler-Maclaurin formula [2]—but they're not necessary in our application because the correction terms vanish anyway.

[221](#)

Although, confusingly, people often still call it a stretched exponential even when the exponent is greater than one. This case should really be called a “squeezed exponential.”

[222](#)

The most general form of linear kernel would be $a_k(\eta) = f(\eta) k$ where $f(\eta)$ is an increasing function of η . However, this form can be turned into the one above by a simple change of variables to $\eta' = f(\eta)$, so in fact we are not losing any generality by assuming $a_k(\eta) = \eta k$.

[223](#)

The calculation is essentially the same as the one leading to Eq. (14.34).

[224](#)

Bianconi and Barabási, who are physicists, solved their model initially by showing that it can be mapped onto the standard physics problem of “Bose-Einstein condensation” in an ensemble of non-interacting bosons. For a physicist already familiar with Bose-Einstein condensation, this provides a quick and elegant way of deriving a solution. For non-physicists, on the other hand, the mapping is probably not very illuminating.

[225](#)

The model is, in this respect, reminiscent of the much simpler and older model of a growing system called *Pólya’s urn*. In this model an urn (i.e., a large pot) initially contains two balls, one green and one red. Repeatedly we draw one ball at random from the urn and replace it with two of the same color. In the limit where the number of balls becomes large, the fraction of green (or red) balls tends to a constant, but the value of that constant is entirely unpredictable—it depends on the details of the fluctuations in the numbers of balls at the early stages of the growth process and all values of the constant are equally likely in the $n \rightarrow \infty$ limit.

[226](#)

The statistics of these leader changes are themselves non-trivial. They obey a so-called *record dynamics*, an interesting non-stationary process that has been studied in its own right, for example by Sibani and Littlewood [296].

[227](#)

We use the word “copying” figuratively here, but in fact there is evidence to suggest that some people really do just copy citations from other papers, possibly without even looking at the cited paper. Simkin and Roychowdhury [297,298] have noted that there is a statistically surprising regularity to the typographical errors people make in citing papers. For instance, many different authors will use the same wrong page number in citing a particular paper, which suggests that rather than copying the citation from the paper itself, they have copied it from an erroneous entry in another bibliography. This does not prove that they did not read the paper in question, but it makes it more likely—if they had actually looked up the paper, there is a good chance they would have noticed that they had the page number wrong.

[228](#)

Kleinberg *et al.* themselves proposed a different model of the copying process in their paper, but their model is quite complex and doesn’t lend itself easily to analysis. The model described here is a simplified realization that possesses the important features of the process while remaining relatively tractable. We note also that Kleinberg *et al.* were not in fact concerned with citation networks in their paper. Their focus was the World Wide Web. We use the language of citation networks here to emphasize the parallels with Price’s model, but the discussion could equally have been framed in the language of the Web.

[229](#)

In simply adding together our probabilities we are technically writing down an expression for the expected number of new edges the vertex receives, rather than the probability of receiving a new edge. However, in the limit of large n the two become the same.

[230](#)

This is a relatively recent development, at least in the United States, where industry regulations made the hub-and-spoke system impractical until 1978. After regulations were lifted the hub-and-spoke system was rapidly adopted by most of the major airlines. Hub-and-spoke systems were also adopted by the package delivery industry around the same time.

[231](#)

The derivation of this result is as follows. If λ is sufficiently large then, as we have shown, the optimal network is the star graph. If we now reduce λ slowly then at some point we enter a regime in which the cost of adding an edge is sufficiently offset by the corresponding reduction in the mean geodesic distance that it becomes worthwhile to add edges between the “spoke” vertices in the star graph. To calculate the point at which such additions become beneficial let us take our star graph and add to it some number r of extra edges. Necessarily these edges fall between the spoke vertices, since there is nowhere else for them fall, and in doing so they form paths of length one between pairs of vertices whose previous shortest path was of length two. The shortest paths between no other vertices are affected by the addition. Thus the total number of vertex pairs connected by paths of length 1 is $n - 1 + r$ and all the rest have paths of length two. Then the mean geodesic distance, as defined in Eq. (7.31) is

$$\ell = \frac{1}{n^2} \sum_{ij} d_{ij} = 2 \frac{(n-1+r) + 2[\frac{1}{2}n(n-1) - (n-1+r)]}{n^2} = 2 \frac{(n-1)^2 - r}{n^2}.$$

(The leading factor of two comes from the fact that the sum over i, j counts each pair of vertices twice.)

Substituting this expression, along with $m = n - 1 + r$, into Eq. (14.148) then gives

$$E = \lambda(n-1+r) + 2(1-\lambda) \frac{(n-1)^2 - r}{n^2} = \text{constant} + \left[\lambda + \frac{2(\lambda-1)}{n^2} \right] r.$$

This will decrease with growing r only if the quantity in square brackets [...] is negative, i.e., if

$$\lambda < \frac{2}{n^2 + 2}.$$

If this condition is satisfied then it becomes advantageous to add edges between the spoke vertices, and to keep on doing so until the network becomes a complete graph, with every vertex connected to every other. Thus there is a discontinuous transition between two behaviors—the star graph and the complete graph—at the point $\lambda = 2/(n^2 + 2)$. Real distribution and transportation networks appear to be in the star-graph regime.

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Strictly it's $\lceil 2m / c \rceil$, where $\lceil x \rceil$ is the smallest integer not less than x .

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In fact, in the original small-world model, as defined by Watts and Strogatz, only one end of each

edge—say the more clockwise end—was rewired and the other left where it was. This, however, results in a model that never becomes a true random graph even when all edges are rewired, as one can easily see, since each vertex is still attached to half of its original edges and hence would have degree at least $\frac{1}{2}c$. In a true random graph there is no such constraint on degrees; vertices can have degrees of any value between zero and $n - 1$. The original model also imposed some other constraints, such as the constraint that no two edges may connect the same vertex pair. This constraint could be imposed in the version we discuss here, although it makes little difference in practice, since the number of such multiedges is of order $1/n$ in the limit of large n and therefore the multiedges make a small contribution to any results if the network is large.

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Equivalently, one could just say that the number of shortcuts added is drawn from a Poisson distribution with mean $\frac{1}{2}ncp$.

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We consider all edges, including the shortcuts, to be the same length, even though the shortcuts are drawn as being longer in figures like Fig. 15.3. We are regarding the network as a purely topological object, not a spatial one.

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Note that the number of shortcuts can be large even when the density of shortcuts remains small, as it must for the scaling form (15.12) to be valid at all.

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In the sociology literature exponential random graphs are also called *p-star models* (sometimes written “*p**”).

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One can also define exponential random graphs models for sets that include non-simple graphs, but the case considered here of simple graphs is the most commonly studied one.

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Sometimes the graph Hamiltonian is defined to be minus this quantity and a corresponding minus sign is introduced in Eq. (15.26). This is by analogy with similar quantities in statistical physics, where the Hamiltonian is an energy function and lower energies correspond to higher probabilities. In studies of networks, however, the definitions are most commonly as given here.

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Those familiar with free energy in its original thermodynamic context may find this expression odd because it varies with network size as n^2 to leading order. In thermodynamic systems, by contrast, free energy is always directly proportional to system size. However the degrees of freedom or “particles” in our network are really the edges (or absence of edges) between vertex pairs, not the vertices themselves, and there are $\binom{n}{2}$ vertex pairs, which is why Eq. (15.40) is proportional to $\binom{n}{2}$.

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We encountered continuous phase transitions previously in Chapters 12 and 13—the point at which the giant component first appears in a random graph is a continuous phase transition, although admittedly it is not obvious that there is a connection between the behavior of the giant component and that seen in Fig. 15.7. The study of phase transitions is an intriguing and beautiful branch of physics that has important implications in areas as diverse as superconductivity, elementary particles, and the origin of the universe. Readers interested in learning more are encouraged to consult, for example, the book by Yeomans [331].

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If you’re interested in the study of percolation in physics the book by Stauffer and Aharony [304]

contains a lot of interesting material on the subject, although most of it is not directly relevant to percolation on networks.

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In most of the physics literature on percolation the occupation probability is denoted p , but we use φ because the letter p is used for many other things in the theory of networks and could cause confusion.

[244](#)

In most of the literature on percolation theory, the giant cluster is called the *spanning cluster*. The reason is that most work on percolation has considered low-dimensional lattices such as the square lattice. On such lattices the giant cluster is distinguished by being the only cluster that spans the lattice from one side to the other in the limit of large n . There is no equivalent phenomenon for percolation on general networks, however, since networks don't have "sides," so the concept of spanning is not a useful one.

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This solution of the percolation problem has a history stretching back some years. In 1961, Fisher and Essam [120] derived a solution for percolation on regular trees (called Cayley trees or Bethe lattices in physics), which is equivalent to the solution given here for the case where every vertex has the same degree. The developments for general degree distributions, however, were not given till some decades later [62,74].

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From Eq. (13.101) we know that a configuration model network has a giant component if and only if $g_1'(1) > 1$, and thus loses its giant component at the point where $g_1'(1) = 1$. Substituting from Eq. (16.13), our network loses its giant component when $2/(e^\lambda - 1) = 1$, i.e., when $\lambda = \ln 3$. See also Problem 13.3 on page 484 for another derivation of this result.

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A phase transition is *continuous* if the quantity of interest, also called the *order parameter* (S in this case), is zero on one side of the transition and non-zero on the other, but its value is continuous at the transition itself. The alternative to a continuous phase transition is a *first-order phase transition*, in which the order parameter jumps discontinuously as it crosses the transition point.

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This excludes the power-law case shown in Fig. 16.3, which is discussed separately below.

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To be more precise, the transition is a *second-order* transition—one where the order parameter is continuous at the transition but its derivative is not.

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If we want to be more careful and keep track of the correction terms we can make use of Eq. (13.51) and integrate Eq. (16.26) to show that $g_0(u) = 1 - \frac{1}{2}k(1-u) + c(1-u)^{\beta+1}/(\beta+1)$. The last term vanishes faster than those before it as $u \rightarrow 1$ because $\beta > 0$ and hence $g_0(u) \simeq 1 - \frac{1}{2}k(1-u)$. This is at first slightly surprising—one would imagine that the correction term ought to be $O(1-u)^2$ —but this type of behavior is common with power-law distributions.

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To the extent that one can regard a power-law network as having a percolation transition at $\varphi = 0$ it is interesting to ask what the order of this transition is. The answer is unclear since Eq. (16.29) doesn't perfectly fit the standard forms for continuous phase transitions. If we define a transition to

be second-order if the order parameter is continuous at the transition and third-order if its derivative is continuous, then the transition is third-order in this case. But one could also argue that the transition is of fractional order between two and three since it varies from zero as a fractional power of the occupation probability ϕ .

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If you look at the epidemiology literature you will sometimes see the infected state referred to as “infective.” There’s no difference between the two terms; they are synonymous. You may also see the word “infectious” used, but this may mean something slightly different. As discussed later in the chapter, more sophisticated models of disease distinguish between a state in which an individual has a disease but it has not yet developed to the point where the individual can pass it on, and a state where they can pass it on. This latter stage is sometimes called the “infectious” stage, a name chosen to emphasize that the disease can be communicated. (The former state is usually called the “exposed” state.) In the present simple two-state model, however, there is no difference between infected and infectious; all individuals who are one are also the other.

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It might be more logical to use $I(t)$ for the number infected, and many authors do so, but we use X instead to avoid later confusion with the index i used to label vertices.

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For convenience we will usually drop the explicit t -dependence of $S(t)$ and $X(t)$ and, as here, just write S and X .

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There aren’t many diseases that really saturate their population like this. Most real diseases that don’t kill their victims are eventually defeated by the immune system. In addition, for many diseases some fraction of the population has a natural immunity that prevents them from being infected (meaning that when exposed to the pathogen their immune system sees it off so quickly that they never become infectious). And some diseases spread so slowly that a large fraction of the population never catches them because they die of other causes first. None of these phenomena is represented in this model.

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This is only approximately true. If people really do have a certain average number of contacts per unit time and assuming those contacts are with living people, then the presence of living but recovered people in the population reduces the number of contacts between infected and susceptible individuals. If, on the other hand, people die rather than recover from the disease then only susceptible and infected individuals are alive and the number of contacts between them will be correspondingly greater. In effect, a person whose acquaintance dies from the disease will (on average) gain one new acquaintance from among the living to replace them, and that new acquaintance might be infected, or might become infected, thereby increasing the chance of transmission of the disease. This effect can easily be incorporated into the model, but we don’t do so here.

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Note that when $\gamma = 0$, as in the SI model, Eq. (17.16) implies that $R_0 \rightarrow \infty$. This is because an infected individual remains infected indefinitely in the SI model and hence can infect an arbitrary number of others, so that R_0 is formally infinite. In any population of finite size, however, the empirical value of R_0 will be finite.

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The solution can be written in closed form using the *Lambert W -function*, which is defined to be the solution of the equation $W(z)e^{W(z)} = z$. In terms of this function, the size of the epidemic is given by

$$S = 1 + \frac{W(-\phi c e^{-\phi c})}{\phi c},$$

Alternatively, we can rearrange Eq. (17.33) to give ϕ as a function of S rather than the other way around:

$$\phi = -\frac{\ln(1-S)}{cS}.$$

This expression can be useful for making plots of S .

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The differences in parameters arise because we are considering a slightly different disease process (one in which each individual is infectious for the same amount of time, rather than the exponential distribution used in the fully mixed model), and also because in the network model β is the transmission rate per edge, rather than the rate for the whole network—this is what gives us the factor of c in the exponent of Eq. (17.33).

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On a finite network with n vertices the equations will in fact close once we get all the way up to combinations of n variables, but this limit is not useful in practice as the equations will become unmanageably numerous and complicated long before we reach it.

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The network was generated using the clustered network model of Ref. [240].

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This equation will diverge if $g_1'(1) = 1$. However, since we are performing the calculation on the giant component of the network, and since the giant component only exists if $g_1'(1) > 1$ —see Section 13.8—we can safely rule out this possibility.

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This contrasts with the approach we took in Section 17.8 where all vertices remained infected for the same amount of time and then recovered. Thus the model studied in this section is not exactly the same as that of Section 17.8, being more similar to the traditional SIR model of Section 17.3. We will see some minor consequences of this difference shortly.

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We can see that the approach of this section cannot be exactly correct from the behavior of Eq. (17.79) on very sparse networks. On a vanishingly sparse network, with only a very few edges and no giant component, κ_1 becomes very small, though still non-zero. On such a network Eq. (17.79) implies that we could, nonetheless, have an epidemic if β is very large or γ very small. Clearly this is nonsense—there can be no epidemic in a network with no giant component. Thus the equation cannot be exactly correct.

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We observe that

$$\frac{d}{dt}(e^{\gamma t} x_k) = e^{\gamma t} \left(\frac{dx_k}{dt} + \gamma x_k \right) = e^{\gamma t} \beta k v s_k,$$

where we've used Eq. (17.82) in the second equality. Integrating and using Eqs. (17.81) and (17.88), we then have

$$x_k(t) = e^{-\gamma t} \left[x_0 + \beta k s_0 \int_0^t e^{\gamma t'} |u(t')|^k \left(1 + \frac{\gamma}{\beta} \ln u(t') - s_0 g_1(u(t')) \right) dt' \right],$$

and $r_k = 1 - s_k - x_k$.

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And like Eq. (17.79) it is also clearly wrong on sparse networks for the same reasons—see footnote 13 on page 664.

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There are also a couple of other rarer possibilities. A fixed point with $\lambda = 0$ can be *neutral*, meaning it neither attracts nor repels. Points near a neutral fixed point stay exactly where they are, meaning that they are fixed points too. For example, the choice $f(x) = 0$ for all x has a neutral fixed point at every value of x . Another less trivial possibility is that a fixed point with $\lambda = 0$ may be of *mixed* type, meaning that it attracts on one side and repels on the other. An example is $f(x) = x^2$ which has a fixed point at $x = 0$ that is attracting for $x < 0$ and repelling for $x > 0$.

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The model we use is a somewhat simplified version of Kleinberg's. His model, for instance, used a two-dimensional lattice instead of a one-dimensional ring as the underlying structure on which the model was built. The calculations, however, work just as well in either case. Our model also places shortcuts at random, where Kleinberg's fixed the number attached to each vertex to be constant and also made them directed rather than undirected.

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The maximum length of a shortcut is $\frac{1}{2}(n-1)$ if n is odd and $\frac{1}{2}n$ if n is even. We will assume that n is odd in this case, which avoids some small annoyances in the derivations.

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Note that both the numerator and denominator of the fraction in Eq. (19.3) vanish at $\alpha = 1$, so one must use l'Hopital's rule to extract the limiting value. The same goes for Eq. (19.8).

[271](#)

In fact, since Kleinberg was studying a two-dimensional version of the small-world model, his result was for $\alpha = 2$, not $\alpha = 1$. In general, on a small-world network built on a d -dimensional lattice, the greedy algorithm succeeds in finding the target in time $O(\log^2 n)$ only when $\alpha = d$ and for all other values takes time increasing at least as a power of n .

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A similar model was also proposed independently by Kleinberg [179].

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Watts *et al.* actually wrote the expression as $Ce^{-\beta m}$, but the difference is only in the value of β and we find the definition (19.9) more convenient.